Finite fields and Chebotarev

Quadratic success and the limits of cubic and fourth-power packing

The arithmetic mechanism proves a fixed power saving for transverse rank-one quadratics. For the audited cubic and fourth-power schemes, the local defect survives but the fully charged rate fails.

PROMINENT PROVED RESULT / Q1-Q2

For fixed integral coefficients with $\gcd(a,b)=1$ and $ab(ae-bc)\neq 0$, the polynomial

$$P(x,y)=(ax+by)^2+cx+ey$$

has arbitrarily large finite $A\subset\mathbb{R}$, using the same set in both coordinates, with

$$|P(A,A)|\le |A|^{2-c_P},\qquad c_P>0 \text{ fixed.}$$

Proved within the supplied source system. Every ordered pair is covered.

The explicit reference model

$$f(x,y)=(x-y)^2+x+y,\qquad f_Q=(x-y)^2+Q(x+y).$$

With fixed split primes, $\rho=\prod_{p\mid Q}(p+1)/(2p)$. Choose $\theta=13D\rho<1$, then fix $X\ge\max\{Q,2\sqrt{D},1\}$. The whole image satisfies

$$|f_Q(B,B)|\le\theta^d|B|^2,\qquad c=\frac{-\log\theta}{\log(2X+1)}>0.$$

Exact scaling gives $A=Q^{-1}\sigma_1(B)$ and $f_Q(Qx,Qy)=Q^2f(x,y)$. This reproduces the polynomial theorem using the original paper's finite-list tower input.

> **The scope is essential.** T01-T04 remain OPEN. A failed upper certificate is not an actual-image lower bound, and polynomial degree alone is not an impossibility criterion.

The gate and the exact limits

CRT gives repeated local restrictions; uniform packing must count every permitted lift. For input radius X, root-discriminant bound D, modulus m, allowed density η, and entire-output radius $H(X)$,

$$\Theta(X) = D\eta\frac{2H(X)+m}{X^2} < 1.$$

Equivalently, lower input rate a and whole-output rate b need $b < 2a$. Input thinning is charged twice. An upper input rate B certifies every strict $c < (2a-b)/B$. [G1-G6]

T01 CUBIC / C5-C7

For $P_3 = (x-y)^3 + x + y$, the exact cube-residue rectangle has

$$\inf_X \beta_p(X) = R\frac{p^2-p+3}{3p^2}(2p+1) > 8R$$

for the recorded $p \equiv 1 \pmod{12}$, with R the tower root discriminant. Additional good primes and usable precision do not repair that specified certificate.

T02 FOURTH POWER / F2-F5

For $P_4 = (x+2y)^4 + 3x - y$, the symmetric certificate has $\Theta_{4,\mathrm{sym}}(X) > 1$ at every radius. The ramified-seven refinement improves the twice-charged local factor only once:

$$\frac{4}{7} \longrightarrow \frac{22}{49}, \qquad N_1 = 4, \quad N_m = 22\,7^{3m-4}\ (m \geq 2).$$

The local count is proved; the audited full-interval global rates remain negative.

NORM-ONE CUBIC GAP / N1-N5

The original directions give bounded real traces T_d, but the needed all-pairs premise remains OPEN:

$$|P_3(T_d, T_d)| \leq Cd^r e^{-\kappa d}|T_d|^2, \qquad \kappa > 0.$$

Growing cubic fibers must connect many distinct products and ratios. Selected translation edges and collision energy do not supply this bound.

Other boundaries. Full-rank binary quadratics have full image at good odd primes [Q3]. A quartic such as f^2 inherits the quadratic saving [F6]. The original finite-list tower and the remarks' infinite common split-prime tower are separate inputs [A5, A8].

READING GUIDE

Results first, proofs and sources after

The positive quadratic theorem comes first. The cubic and quartic limits, and then the norm-one support gap, follow. The arithmetic assembly, common counting lemmas and detailed provenance remain available in the later chapters and appendices.

1 The quadratic theorem that works	PROVED / FULL CARTESIAN SUPPORT
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Appendix D Claim records and source identities	43 INDEXED RECORDS / 23 LOCAL SOURCE IDENTITIES

Reading the status labels. PROVED refers to a complete identified argument in the source system, with its inspection scope stated. EXTERNAL RESULT names an imported theorem. FAILED describes a specified certificate or carrier. OPEN preserves an unresolved construction or support premise. Reported C1 statements whose original proofs are unavailable are marked explicitly.

The source report contains 43 indexed statements and source records. Original theorem numbers and local report IDs remain distinct. All same-set witnesses count every ordered pair, and all finite parameters precede the growing field degree.

Fixed targets kept distinct. To1: $(x - y)^3 + x + y$; To2: $(x + 2y)^4 + 3x - y$; To3: $x^2 + xy + y^2$; To4: $(x - y)^3 - 3(x + y)$. Each remains OPEN. Positive rank-one quadratics and their quartic compositions are separate results.

PROVED / FULL CARTESIAN SUPPORT

1 The quadratic theorem that works

Theorem Q1 The transverse rank one family

PROVED within the supplied source system; complete argument inspected and reproduced. Let

$$L = ax + by, \qquad T = cx + ey,$$

have integral coefficients with $\gcd(a, b) = 1$ and $ab(ae - bc) \neq 0$. For the fixed polynomial $P = L^2 + T$, there are arbitrarily large finite $A \subset \mathbb{R}$ and a fixed $c_P > 0$ with

$$|P(A, A)| \leq |A|^{2-c_P}.$$

Proof. Put $A_0 = (|a| + |b|)^2$, $A_1 = |c| + |e|$, and $C_P = 2(A_0 + A_1) + 1$. The original tower input A5 supplies $t \asymp \ell^2$ primes $p_i \equiv 1 \pmod 4$, with $\log D = O(\ell \log \ell)$. Choose ℓ large enough that

$$C_P D (3/5)^t < 1.$$

After the primes are selected, fix $Q = \prod_i p_i$, $\rho = \prod_i (p_i + 1)/(2p_i)$, and $X > \max\{Q, \sqrt{D}, 1\}$. All these parameters precede the growing tower layer.

Use $F_Q = L^2 + QT$. Since L is primitive it is surjective over every selected residue field, so every output is a square in each CRT coordinate. G4 gives at most $(\rho Q)^d$ classes. Every conjugate of every output has absolute value at most $A_0 X^2 + A_1 Q X$. G5 therefore gives

$$|F_Q(B_K(X), B_K(X))| \leq \theta^d |B_K(X)|^2, \qquad \theta = C_P D \rho < 1.$$

The embedding and common input scaling $A = Q^{-1} \sigma_1(B_K(X))$ preserve both cardinalities by G7. With $n \leq (2X + 1)^d$, choose

$$c_P = -\log \theta / \log(2X + 1) > 0.$$

The lower input count tends to infinity, proving the fixed-polynomial, same-set theorem. Every ordered pair was counted, including the diagonal. □

For $P_* = (x + 2y)^2 + 3x - y$, the sharper displayed certificate is

$$\Theta_*(X) = D\rho(18 + 8Q/X + Q/X^2) < 27 D \rho.$$

The original finite-list tower is sufficient. An infinite common split-prime set, regulator estimate and class-group selection for the input carrier are unnecessary here. The class-group bound is an ingredient in the original unit-distance direction count, not a further charge in this quadratic box proof.

Nonspecialness under the stated hypotheses. An additive representation $g(u(x) + v(y))$ of total degree two either has linear g, which cannot produce a mixed term, or quadratic g with affine inner functions. In the latter case its linear part is parallel to its rank-one quadratic direction, requiring $ae - bc = 0$. A multiplicative representation at degree two has affine g, u, v, and its leading part is a multiple of xy, whereas L^2 has nonzero x^2 and y^2 terms. Hence the family is nonspecial. The conditions $ab \neq 0$ and transversality qualify this claim; they must not be omitted from the sourced theorem.

Source: corrected C0, Theorem C0.1 and §§4-6, dated 2026-09-03. External novelty has not been assessed by this report.

Theorem Q2 The reference quadratic with the factor 13

EXTERNAL RESULT and independently reconstructed elementary implication. For

$$f = (x-y)^2 + x + y, \qquad f_Q = (x-y)^2 + Q(x+y),$$

choose fixed split primes so that $\theta = 13D\rho < 1$, then fix $X \geq \max\{Q, 2\sqrt{D}, 1\}$. The centered box gives

$$(X/\sqrt{D})^d \leq n \leq (2X+1)^d, \quad |f_Q(B,B)| \leq (13D\rho)^d n^2.$$

The normalized real set $A = Q^{-1}\sigma_1(B)$ therefore satisfies

$$|f(A,A)| \leq n^{2-c}, \qquad c = -\log(13D\rho)/\log(2X+1) > 0.$$

Proof of the numerical constant. All outputs have conjugate radius at most $2QX + 4X^2 \leq 6X^2$. G3-G4 yield $[\rho(12X^2+Q)]^d \leq (13\rho X^2)^d$. Divide by the squared input lower bound. The identity $f_Q(Qx, Qy) = Q^2 f(x,y)$ supplies the exact common transport. □

For every fixed $\epsilon > 0$, choosing larger fixed X gives $|A+A| \leq |A|^{1+\epsilon}$: use $B+B \subset B_K(2X)$, the upper bound $(4X+1)^d$, and choose $4X+1 \leq (X/\sqrt{D})^{1+\epsilon}$. The saving exponent may decrease but remains positive.

This is Pohoata, *Split primes and the Elekes-Rónyai problem*, arXiv:2606.13619v2, Theorem 1.1, Lemmas 3.2-3.3 and Proposition 4.1. His Proposition 3.1 uses the stronger infinite common split-prime tower. Q2 also follows from the original finite-list input A5, because $13D\rho < 1$ can be arranged before the primes and box radius are fixed. The original unit-distance paper itself does not state Q2 or the constant 13.

LOCAL COUNT PROVED / CERTIFICATE FAILS

2 Cubic defects and the packing obstruction

Cubic local image

C1 - PROVED; inspected elementary argument. For every odd prime p, integer C, and $k \geq 1$, set $r = \min(v_p(C), k)$, taking $v_p(0) = \infty$. On $R_k = \mathbb{Z}/p^k\mathbb{Z}$, the full image of $P_C(x, y) = (x - y)^3 + C(x + y)$ is exactly

$$\{z \in R_k : z \bmod p^r \text{ is a cube}\}.$$

If $N_r(a)$ counts cube roots modulo p^r, its ordered fiber size is

$$r_{P_C,k}(z) = p^k N_r(z \bmod p^r).$$

For $r = 0$, the quotient has one class and $N_0 = 1$. Source: cubic checkpoint §E.1; polynomial and quantifiers are as stated.

Proof. The invertible change $d = x - y$ gives $P_C = d^3 + Cd + 2Cy$. Since 2 is a unit, $2Cy$ ranges over $p^r R_k$, proving the image statement. For a permitted z, there are $N_r(z)p^{k-r}$ choices of d; for each, the equation for y has p^r solutions. Their product is the claimed fiber size. This also covers $r = k$, where C vanishes in R_k.

C2 - PROVED; inspected count. For odd $p \neq 3$, put $g = \gcd(3, p - 1)$. For every $r \geq 1$, the number of cubes is

$$c_p(r) = 1 + \frac{1}{g} \sum_{h=0}^{\lfloor (r-1)/3 \rfloor} \varphi(p^{r-3h}).$$

In particular, for $p \equiv 1 \pmod 3$,

$$\delta_{p,r} := \frac{c_p(r)}{p^r} = p^{-r} + \frac{p - 1}{3p} \sum_{h=0}^{\lfloor (r-1)/3 \rfloor} p^{-3h},$$

$$\delta_{p,1} = \frac{p + 2}{3p}, \qquad \delta_{p,2} = \frac{p^2 - p + 3}{3p^2}, \qquad \lim_{r \to \infty} \delta_{p,r} = \frac{p^2}{3(p^2 + p + 1)}.$$

Source: §E.1; dependency C1 for application to P_C.

Proof. A nonzero cube has valuation $3h < r$. On units modulo p^s, cubing has kernel size g: there are g cube roots of unity in $\mathbb{F}_p$, and each lifts uniquely because its derivative $3x^2$ is a unit. Thus the valuation-$3h$ shell contains $\varphi(p^{r-3h})/g$ cube classes. Add the zero class and divide by p^r. The limit follows by summing a geometric series. This formula does not cover $p = 3$; C1's image description still does.

C3 - PROVED conditional on the stated splitting; inspected arithmetic. If p splits completely in a degree-d number field L, then

$$\mathcal{O}_L/p^k\mathcal{O}_L \cong (\mathbb{Z}/p^k\mathbb{Z})^d.$$

Indeed every completion at a prime above p is $\mathbb{Q}_p$; the prime-power Chinese remainder decomposition gives the identity. Consequently C1 imposes exactly d copies of its scalar restriction, with density $\delta_{p,r}^d$. Distinct rational primes multiply by CRT. This is a product-ring identity, not a random independence assumption. Sources: §E.1 and the identified tower input.

For the fixed-polynomial clearing identity

$$M^3 P_3(u/M, v/M) = (u-v)^3 + M^2(u+v), \tag{C.1}$$

usable precision at $p \mid M$ saturates at $2v_p(M)$. Beyond this, C1 gives the inverse image of the same cube set, so its density no longer decreases. Complete splitting amplifies this fixed defect vertically; it does not turn increasing precision into an unbounded local gain.

Cubic range and population

C4 - PROVED; inspected exact continuous range. For real $I = [c - X/2, c + X/2]$, $X > 0$, and integer $M > 0$, the image of $Z_M(u,v) = (u-v)^3 + M^2(u+v)$ on I^2 is an interval of width

$$W_M(X) = 2\max(X^3, M^2 X). \tag{C.2}$$

Source: §E.2 and §H.4; the polynomial has the positive linear sign.

Proof. Put $d = u - v$, $s = u + v - 2c$. Their feasible region is $|d| + |s| \le X$, and the centered output is $d^3 + M^2 s$. For $|d| = t$, its largest absolute value is $t^3 + M^2(X - t)$. This convex function has its maximum at $t = 0$ or $t = X$. The region is centrally symmetric, both extreme signs occur, and its continuous image is connected. Translation changes only the output center $2M^2 c$.

C5 - PROVED upper certificate; inspected full-pair counting. Let L be totally real of degree d, root discriminant R, and let a fixed prime $p \equiv 1 \pmod 3$ split completely in L. Averaging gives a translate of a side-X Minkowski cube whose integral section B satisfies

$$(X/\sqrt{R})^d \le |B| \le (X+1)^d.$$

For $A = \sigma_1(B)/p$, every ordered pair satisfies

$$|P_3(A,A)| \le \left[\delta_{p,2}\{p^2 + 2\max(X^3, p^2 X)\}\right]^d. \tag{C.3}$$

Sources: §§D.2, E.3 and F.1; dependencies C1-C4, lattice covolume, and the integral norm. All primes, denominators and widths are fixed before tower degree grows; $X > \sqrt{R}$ supplies input growth.

Proof. The lattice covolume is $R^{d/2}$, giving the lower count by translated averaging. A nonzero algebraic integer has norm of absolute value at least one, so two integral points differ by at least one in some real coordinate; packing side-one cubes proves the upper count. The real field embedding and division by p are injective.

The cleared output in (C.1) occupies at most $(p^2\delta_{p,2})^d$ residue classes modulo $p^2\mathcal{O}_L$. Within each class, differences are $p^2\gamma$, with integral $\gamma \neq 0$; one conjugate difference is at least p^2. Packing side-p^2 cubes in the output rectangle bounds each class by $(1 + W_p(X)/p^2)^d$. Multiplication gives (C.3). No input equidistribution or omitted-pair hypothesis is needed.

Dividing this certificate by the squared input lower bound yields

$$\beta_p(X) = R\delta_{p,2}\left[2\max(X, p^2/X) + p^2/X^2\right]. \tag{C.4}$$

The root-discriminant cost, fixed denominator and cubic height are all present. Class groups and regulators do not enter this box calculation.

C6 - PROVED optimization; FAILED certificate. For every eligible $p \geq 7$,

$$\inf_{X>0} \beta_p(X) = R\delta_{p,2}(2p+1). \tag{C.5}$$

For the checkpoint's $p \equiv 1 \pmod{12}$, hence $p \geq 13$,

$$\inf_{X} \beta_p(X) \geq 1431R/169 > 8R \geq 8.$$

Source: §§D.1 and F.2; dependencies C2 and C5.

Proof. For $X \leq p$, the bracket in (C.4) is $2p^2/X + p^2/X^2$, which decreases. For $X \geq p$, it is $2X + p^2/X^2$, whose derivative $2 - 2p^2/X^3$ is positive. The minimum is therefore at $X = p$. The resulting scalar factor is $2p/3 - 1/3 + 5/(3p) + 1/p^2$, increasing for $p \geq 7$. Its value at 13 is $1431/169$. Restricting to $X > \sqrt{R}$ can only increase this lower bound.

The congruence $1 \bmod 12$ comes from retaining splitting in $\mathbb{Q}(i)$ and also prescribing splitting in $\mathbb{Q}(\zeta_3)$, a stated adaptation of the source Chebotarev step. Cube sparsity alone needs $1 \bmod 3$. Even allowing $p = 7$ leaves the base $225R/49 > 1$. Thus the extra congruence is not the reason the certificate fails. A failed upper certificate is not an actual-image lower bound.

Further precision and input restrictions

C7 - PROVED obstruction for the specified packing architecture. Fix

$$M = \prod_{b=1}^{t} p_b^{a_b}, \quad m = \prod_{b=1}^{t} p_b^{r_b} \mid M^2, \quad 1 \leq r_b \leq 2a_b,$$

with distinct completely split primes $p_b \equiv 1 \pmod{3}$, $a_b \geq 1$, and $t \geq 1$. Put $\delta_m = \prod_b \delta_{p_b,r_b}$. For translated anisotropic Minkowski rectangles of half-widths $h_i > 0$, cleared outputs counted by their cube classes and uniform norm packing have normalized certificate

$$R^d \delta_m^d \prod_i \Phi_{M,m}(h_i), \qquad \Phi_{M,m}(h) = \frac{m}{4h^2} + \max(4h, M^2/h).$$

Every coordinate has exact minimum

$$\inf_{h>0} \Phi_{M,m}(h) = 2M + m/M^2.$$

Sources: §§F.2-F.3 and H.4; dependencies C2-C5.

Proof. Use (C.2) with $X = 2h$, modulus m, and divide the population bound by input volume squared. Below $h = M/2$ both terms decrease; above it, the derivative is $4 - m/(2h^3) > 0$, since $m \leq M^2$ and $M \geq 7$. Also

$$p_b^{a_b} \delta_{p_b,r_b} \geq p_b^{a_b} \frac{p_b - 1}{3p_b} \geq 2,$$

so $M\delta_m \geq 2^t$. The best normalized base is consequently larger than $2RM\delta_m \geq 2^{t+1}R > 1$. Anisotropy and translation do not repair it. Beyond coefficient saturation, the density is unchanged while a larger modulus increases the same population expression; extra digits do not improve this certificate either.

The inherited algebraic-denominator extension is also scoped: take $\mu \in L^\times$ with $\mu^2 \in m\mathcal{O}_L$, where m is the same completely split good-prime modulus specified above, and put $a_i = |\sigma_i(\mu)|$. A denominator in an extension of L would require its own field degree and embedding ledger. Its normalized factors obey

$$\frac{m}{4h_i^2} + \max(4h_i, a_i^2/h_i) \geq 2a_i, \qquad |N(\mu)| \geq m^{d/2}.$$

Their geometric mean is at least $2\sqrt{m}$. For good primes, $\sqrt{p^r}\delta_{p,r} > 1$: for $r = 1$ this is $(p+2)/(3\sqrt{p}) > 1$; for $r \geq 2$ the bound is at least $(p-1)/3 \geq 2$. Thus this variant also has base greater than one. This closes the displayed denominator-and-rectangle argument, not arbitrary algebraic input sets.

The same source §F.3 records three exceptional-prime tests. Even granting complete splitting at 2 and 3, which the supplied tower does not assert, their optimized certificate bases are

Restriction	Optimized base
Unscaled cubic parity	$(3/2)2^{1/3}R$
Denominator 3, cubes modulo 9	$7R/3$
Both preceding restrictions	$4R/3$

These are additional inspected deductions under C7's **FAILED certificate** scope, rather than claims that 2 or 3 can be used in the existing tower. To verify them, write the unscaled polynomial as $d^3 + d + 2y$. Since $d^3 + d$ is even and $d = 0$ reaches every even residue, its image modulo 2^k is exactly the even classes. The density stays $1/2$; the best packing modulus is 2. The normalized factor is $R[X^{-2} + \max(X, X^{-1})]$. On $X \leq 1$ it decreases; on $X \geq 1$ its derivative vanishes at $X = 2^{1/3}$, giving the first base.

For denominator 3, the cubes modulo 9 are $0, 1, -1$. The factor is $(R/3)[9/X^2 + 2\max(X, 9/X)]$, decreasing to $X = 3$ and increasing thereafter, giving $7R/3$. The cleared polynomial also has the parity restriction. Combining it with the modulo-9 condition uses modulus 18, density $1/6$, and factor $(R/6)[18/X^2 + 2\max(X, 9/X)]$, likewise minimized at 3. All three bases exceed one. This paragraph settles only these explicitly specified exceptional restrictions; it is not a theorem about every mixed-prime coefficient or every restricted input set.

C8 - PROVED obstruction with input thinning charged twice. Let $M = \prod_{b=1}^{t} p_b$ be squarefree as above, $\mathcal{R} = \prod_b \mathbb{F}_{p_b}$, and let an arbitrary nonempty correlated code $\mathcal{C} \subset \mathcal{R}^d$ restrict inputs. Write $\mathcal{D} = (\mathcal{C} - \mathcal{C})^{\circ 3}$. The averaged-input/uniform-lift certificate still fails. Source: §E.4 and export §6.3; dependencies C4-C5 and elementary finite-ring fibers.

Proof. A prescribed output difference-cube has at most 3^t differences per coordinate, each with exactly M ambient ordered pairs. Therefore

$$|\mathcal{D}| \geq |\mathcal{C}|^2/(3^t M)^d.$$

Translated averaging guarantees only $|B| \geq |\mathcal{C}|X^d/(M^d R^{d/2})$. Counting the allowed output classes uniformly gives the normalized certificate

$$\frac{M^d|\mathcal{D}|}{|\mathcal{C}|^2}\left[\frac{R\{M+2\max(X^3,M^2X)\}}{X^2}\right]^d.$$

Its base is at least $3^{-t}R(2M+1/M)>2R\prod_b(p_b/3)>1$. The squared denominator $|\mathcal{C}|^2$ is essential. This does not exclude an anomalously dense input translate combined with a new residue-sensitive output count, or arbitrary sparse real carriers.

Continuous positive-volume covers

C9 - PROVED continuous-cover theorem; inspected complete argument. Let $m\geq 1$, let $W\subset\mathbb{R}^m$ be bounded Borel with positive volume V, and define the coordinatewise full image

$$\mathcal{P}_m(W)=\{(x-y)^{\circ 3}+x+y : x,y\in W\}.$$

If N_1 denotes the least number of translates of half-open unit axis cubes covering this image, then

$$N_1(\mathcal{P}_m(W))>(\sqrt{3}/\pi)^m V^2. \tag{C.6}$$

Source: the 3 September continuous checkpoint §F.3, equations (F.6)-(F.10); dependency only the displayed analytic argument.

Proof. Fix any $y\in W$. The map $\Phi_y(x)=(x-y)^{\circ 3}+x+y$ is a global continuously differentiable diffeomorphism: each coordinate has positive derivative $1+3(x_i-y_i)^2$ and ranges over the real line. Its Jacobian is $J_y(x)=\prod_i(1+3(x_i-y_i)^2)$. Change of variables and Cauchy-Schwarz give

$$V^2\leq\left(\int_W J_y\right)\left(\int_W J_y^{-1}\right)=\mathrm{vol}(\Phi_y(W))\int_W J_y^{-1}.$$

Since W is bounded and the integrand is everywhere positive,

$$\int_W J_y^{-1}<\int_{\mathbb{R}^m}J_y^{-1}=\left(\int_{\mathbb{R}}\frac{dt}{1+3t^2}\right)^m=(\pi/\sqrt{3})^m.$$

Thus $\mathrm{vol}(\Phi_y(W))>(\sqrt{3}/\pi)^m V^2$. Every cover of the full image covers this fixed-input slice; a union of N unit cubes has volume at most N. This proves (C.6) without needing measurability of the entire two-input image.

For the source's volume/covolume input count followed by a full continuous-image cover and norm packing, the block factor is

$$\eta_{\mathrm{block}}=R^m N_1(\mathcal{P}_m(W))/V^2>(R\sqrt{3}/\pi)^m.$$

The standard Minkowski discriminant bound gives $R\geq 2$ for every totally real field of degree $d\geq 2$. Precisely, $R\geq(d^d/d!)^{2/d}$; for $d=2$ this lower bound is 2, and for $d\geq 3$, AM-GM gives $(d!)^{1/d}\leq(d+1)/2\leq d/\sqrt{2}$. Consequently $\eta_{\mathrm{block}}>1$. Source: §G.1; this last step uses Minkowski's standard theorem as an identified external input.

This excludes the stated certificate for every bounded positive-volume Borel window, including disconnected, correlated and nonrectangular ones. Its hypotheses concern continuous volume and the cover used to count all permitted lifts. It supplies no lower bound on an arbitrary sparse finite lattice image. A discrete carrier with an additional exact-support argument is outside this theorem's conclusion.

LOCAL IMPROVEMENT / EXACT SATURATION

3 Fourth powers and higher precision

F1 fourth power values and exact fixed polynomial transport

PROVED, elementary identities checked here. For an odd prime p, cyclicity of $\mathbf{F}_p^\times$ gives

$$|\{z^4 : z \in \mathbf{F}_p\}| = 1 + \frac{p-1}{\gcd(4, p-1)}.$$

Thus primes $p \equiv 1 \pmod 4$ give $(p+3)/4$ values, including zero, and density $(p+3)/(4p)$. At $p \equiv 3 \pmod 4$ the density is only the square density $(p+1)/(2p)$. Prime two supplies no such saving. Since $L(x,y) = x + 2y$ is surjective on full field inputs, these are also the exact values of L^4.

Write $T(x,y) = 3x - y$. For a fixed product of selected split primes $Q = \prod_{i=1}^{t} p_i$, the removable deformation is

$$F_Q = L^4 + Q^3 T, \qquad F_Q(Qx, Qy) = Q^4 P_4(x,y). \tag{F1.1}$$

The coefficient is Q^3, not Q. At each selected prime dividing Q, F_Q reduces to L^4. If p_i splits completely in a totally real degree-d field K, then $\mathcal{O}_K/p_i\mathcal{O}_K \cong \mathbf{F}_{p_i}^d$. CRT therefore allows at most

$$(Q\rho_4)^d \quad \text{output classes modulo } Q\mathcal{O}_K, \qquad \rho_4 = \prod_{p \mid Q} \frac{p+3}{4p}. \tag{F1.2}$$

This is quotient-ring product factorization, not probabilistic independence or equidistribution of a bounded carrier. For a finite algebraic carrier C, choose one real embedding σ and set $A = \sigma(C)/Q$. Then (F1.1) preserves input and output cardinalities exactly. Fix the tower, its root discriminant, the primes, Q, and the real radius before the degree grows. Chebotarev in one finite extension alone does not ensure splitting in every tower layer.

F2 the symmetric fourth power certificate has the wrong rate

PROVED certificate obstruction; the actual image is OPEN. Let $\mathrm{rd}(K) = D$, and use $C = B_K(X) = \{u \in \mathcal{O}_K : |\sigma_i(u)| \le X\}$. The half-box translate-and-difference argument and norm separation give

$$(X/\sqrt{D})^d \le n := |C| \le (2X+1)^d.$$

Every conjugate of every output of F_Q has absolute value at most

$$M_4 = 81X^4 + 4Q^3X.$$

This is a valid enclosure obtained by separate estimates, not an exact continuous radius. Distinct outputs in one class modulo $Q\mathcal{O}_K$ are Q-separated in sup norm: their difference is Q times a nonzero algebraic integer. Cubes of side Q consequently bound each class population by $(1 + 2M_4/Q)^d$. Combining this uniform bound with (F1.2) yields

$$|F_Q(C,C)| \le [\rho_4(2M_4 + Q)]^d \le \Theta_{4,\mathrm{sym}}(X)^d n^2,$$

$$\Theta_{4,\mathrm{sym}}(X) = D\rho_4 \left(162X^2 + \frac{8Q^3}{X} + \frac{Q}{X^2}\right). \tag{F2.1}$$

For every $X > 0$, put $z = X/Q$. If $z \geq 1/4$, then $162z^2 \geq 162/16 > 10$; if $z < 1/4$, then $8/z > 32$. Hence the first two terms in parentheses exceed $10Q^2$. Meanwhile

$$\rho_4 Q^2 = \prod_{p \mid Q} \frac{p(p+3)}{4} \geq 10^t,$$

because the selected primes are $1 \pmod 4$ and at least five. Thus $\Theta_{4,\mathrm{sym}}(X) > 1$. No radius makes this particular fully charged comparison prove compression. Adding those primes strengthens the local defect but increases the coefficient/modulus cost even faster.

The conclusion concerns this numerical upper certificate. It does not say the actual image equals the enclosure, give an actual-image lower bound, or exclude a sharper count of globally realized outputs on the same carrier.

F8 Stronger C1 statements and the correlated input boundary

PROVED as reported in the controlling state; original main proofs unavailable for independent inspection in this report. T02's state, §4 C1, records the following stronger results for the same deformation F_Q. For the centered interval $[-X, X]$, writing $z = X/Q$, its exact continuous width is $W(X) = Q^4 w(z)$, where

$$w(z) = \begin{cases} 8z, & 0 < z \leq 40^{-1/3}, \\ 80z^4 + 6z, & 40^{-1/3} \leq z \leq 1/2, \\ 81z^4 + \frac{11}{2}z + \frac{3}{16}, & z \geq 1/2. \end{cases}$$

For every $Q \geq 5$, center $b \in \mathbb{R}$ and radius $r > 0$, it records

$$W_Q(b,r) \geq 7Q^3 r, \qquad W_Q(b,r) \geq (7r/3)^4, \qquad W_Q(b,r) + Q \geq Q(2r+1)^2.$$

For every prime $p \equiv 1 \pmod 4$ and nonempty $E \subseteq \mathbb{F}_p$, the recorded local theorem is

$$\frac{p\,|\{(x+2y)^4 : x, y \in E\}|}{|E|^2} \geq \frac{p+3}{4p},$$

with equality exactly for $E = \mathbb{F}_p$. Thus independent coordinate input restrictions cannot improve that normalized factor after both input losses. The reported sharper interval certificate has

$$\Theta_{4,\mathrm{int}}(X) = D\rho_4 \frac{W(X) + Q}{X^2} > 9(80/3)^{1/3} D\rho_4 Q^2 > 1.$$

Its translated/anisotropic full-interval envelope obeys $U = \rho_4^d \prod_i (W_i + Q)$ and $U/|C|^2 \geq (\rho_4 Q)^d \geq 2^{td}$. The main C1 proof is identified by the recovery package but was not received locally. These precise statements are recorded as reported results, not substituted for the inspected proofs of F2-F5.

PROVED; complete available proof for the correlated finite case. The located `phase_c2/C1_STATUS_ADDENDUM.md`, §2, proves that for every nonempty $S \subseteq \mathbb{F}_5^2$, with $n = |S|$,

$$H_5(S) = \{(x+2y)^{\circ 4} : x, y \in S\}, \qquad \frac{25|H_5(S)|}{n^2} \geq \frac{4}{25},$$

with equality only for $S = \mathbb{F}_5^2$.

Proof. If $n \le 12$, nonemptiness gives the strict lower bound $25/144 > 4/25$. If $n \ge 13$, the sets S and $z - 2S$ intersect for every z in the group of order 25, so $S + 2S = \mathbb{F}_5^2$. Coordinatewise fourth powers then give exactly $\{0,1\}^2$, with four values. The normalized factor is $100/n^2 \ge 4/25$, with equality only at $n = 25$. $\square$

OPEN. The corresponding all-prime/all-dimension inequality

$$\frac{p^d |\{(x+2y)^{\circ 4} : x, y \in S\}|}{|S|^2} \ge \left(\frac{p+3}{4p}\right)^d, \qquad \varnothing \ne S \subseteq \mathbb{F}_p^d, \quad p \equiv 1 \pmod 4,$$

is not established by the finite case. Even a useful opposite defect would still require a compatible real realization and a positive fully charged global rate.

The same available addendum, §1, explains why a weak exponential deficit relative to a full class container can follow from the trivial pair bound: for $r > \max(Q, \sqrt{D})$, its reported width lower bound makes the container base larger than $(2r+1)^2$. This weak deficit supplies no proof that the saving exceeds $\log \Theta_{4,\mathrm{int}}$. An improvement measured against an oversized envelope must cross the actual rate threshold before it certifies compression. No enumeration is a premise of these conclusions.

F7 the C2 reduction and C3 universal one level optimum

PROVED in the identified source system; the complete analytic proof was read and its case argument checked here. In the characteristic-seven ring $R = \mathbf{F}_7[\pi]/(\pi^3)$, assume $7 \nmid Q$ and put $s = Q^3 \bmod 7 \in \{1, -1\}$. Since $L + 2T = 7x = 0$, $L = -2T$, so for **every nonempty** $E \subseteq R$,

$$F_Q(E, E) = h_s(3E - E), \qquad h_s(z) = 2z^4 + sz. \tag{F7.1}$$

This is an exact quotient-ring identity, not a characteristic-zero identity. C2's integral determinant-seven graph encodes the same relation: for a unit u, the map $(t_1, t_2) \mapsto (ut_1 + 2t_2, 3ut_1 - t_2) = (a, b)$ has image exactly $a + 2b \in 7\mathcal{O}$, of index 7^d. The graph index and its scalar consequence h_s are not independent savings.

Expansion gives

$$h_s(a + b\pi + c\pi^2) = h_s(a) + (a^3 + s)b\pi + [(a^3 + s)c + 5a^2b^2]\pi^2. \tag{F7.2}$$

The residue map is a permutation. The three critical residues are $K_1 = \{3, 5, 6\}$, $K_{-1} = \{1, 2, 4\}$, defined by $a^3 = -s$. Each of the other four branches is bijective in its two nilpotent digits. Each critical branch keeps only the square of the first digit. Disjoint output residues therefore give $|h_s(R)| = 4 \cdot 49 + 3 \cdot 4 = 208$.

Charge both input fractions by setting $\Phi_s(E) = 343|h_s(3E - E)|/|E|^2$. C3 proves all three sharp statements:

$$\min_{\varnothing \ne E \subseteq R} \Phi_s(E) = 4/7, \tag{F7.3}$$

with equality exactly at the three full critical fibers $a + \pi R$;

$$\min_{E \text{ meets at least two residue fibers}} \Phi_s(E) = 208/343, \tag{F7.4}$$

with equality only at $E = R$; and, excluding the three extremizers,

$$\min \Phi_s(E) = 343/576, \tag{F7.5}$$

with equality exactly at a full critical fiber minus one point.

Here is the complete case structure. Write $n = |E|$. In a single regular input fiber, fixing one input gives $|3E - E| \geq n$, and the corresponding argument fiber is regular because $2^3 = 1$; thus $\Phi_s(E) \geq 343/n \geq 7$. In a single critical fiber, let $T_1 \subseteq \mathbf{F}_7$ be its first nilpotent-digit support, of size t_1. Then $n \leq 7t_1$, and the output count is exactly $|\{b^2 : b \in 3T_1 - T_1\}|$. Cauchy-Davenport, in the precise form $|U + V| \geq \min(7, |U| + |V| - 1)$, and the at-most-two fibers of squaring give at least $\min(t_1, 4)$ outputs. For $t_1 \leq 6$ the resulting normalized cost is at least $7/9$. For $t_1 = 7$, it is exactly $1372/n^2$, minimized only at $n = 49$.

For a multifiber set, every input residue touches a regular argument branch. A regular residue uses the diagonal argument $2a$. For a critical a and another residue b, if both $3a - b$ and $3b - a$ were critical, putting $t = b/a$ would give $(3 - t)^3 = (3t - 1)^3 = 1$. Their difference is $t - t^2 = 0$, whose two possibilities either fail the first equation or contradict $a \neq b$. On a regular branch reached from input fibers of sizes n_a, n_b, fixing either input gives at least $\max(n_a, n_b)$ distinct outputs. Summing these disjoint regular branches gives at least $n/3$: with two or three input residues use a largest fiber; with four, Cauchy-Davenport makes all seven argument residues occur, so use its contribution and the three remaining branches: writing M, m for the largest and smallest fiber sizes, their contribution is at least $M + 3m$, while $n \leq 3M + m$, so it is at least $n/3$. With at least five a largest fiber touches at least $k - 3 \geq k/3$ regular branches.

For $n \leq 171$ this yields $\Phi_s(E) \geq 343/(3n) > 208/343$. For $n \geq 172$, the two size-n sets $3E$ and $z + E$ intersect for every $z \in R$, so $3E - E = R$ and $\Phi_s(E) = 343 \cdot 208/n^2 \geq 208/343$, with equality only at $n = 343$. This proves (F7.3)-(F7.4). Outside their extremizers, the only remaining case able to attain (F7.5) is a critical fiber with all seven first digits and $n \leq 48$: $1372/48^2 = 343/576$, attained by every one-point deletion. For complete unions $E = A + \pi R$, the source also gives the exact formula $|F_Q(E, E)| = 49r + 4c$, where r, c count regular and critical residues in $3A - A$, and $\Phi_s(E) = (49r + 4c)/(7|A|^2)$. The gap above $4/7$ is exactly $97/4032$. Moreover $\Phi_s(E) < 208/343$ forces 48 or 49 points in one critical fiber, since $1372/47^2 > 208/343$. These are exact discrete stability statements, not asymptotic hypotheses.

These analytic arguments cover all subsets; the historical structured screen is not their proof. No screen or supplied program was rerun here. The sharp minimum applies to the 343-element ring only: F3's higher-level full-fiber count improves it once, without proving any universal higher-level optimum.

F3 the complete C4 local theorem and its saturation proof

PROVED in the identified source system; its analytic argument is inspected and checked below. C4 uses a separate ramified-seven refinement with $Q = 1$. Let $\mathcal{O}$ be the valuation ring of the actual local cubic, with maximal ideal $\mathfrak{p}$, ramification index three and residue degree one:

$$7\mathcal{O} = \mathfrak{p}^3, \quad \mathcal{O}/\mathfrak{p} = \mathbf{F}_7, \quad |\mathcal{O}/\mathfrak{p}^r| = 7^r.$$

For $\alpha \in \{3, 5, 6\}$, so $\alpha^3 = -1 \bmod \mathfrak{p}$, set $\mathcal{E} = \alpha + \mathfrak{p}$, $R_m = \mathcal{O}/\mathfrak{p}^{3m} = \mathcal{O}/7^m\mathcal{O}$, and $E_m = \mathcal{E} \bmod \mathfrak{p}^{3m}$. Then $|R_m| = 7^{3m}$, $|E_m| = 7^{3m-1}$, and

$$N_m := |P_4(E_m, E_m)| = \begin{cases} 4, & m = 1, \\ 22\,7^{3m-4}, & m \geq 2. \end{cases} \tag{F3.1}$$

To prove this for the entire Cartesian image, use the bijection $(x,y)\mapsto(L,x)$, $L=x+2y$, from $\mathcal{E}^2$ to $(3\alpha+\mathfrak{p})\times(\alpha+\mathfrak{p})$. The inverse is $y=(L-x)/2$; two is a unit. Over the actual valuation ring,

$$P_4(x,y)=g(L)+\tfrac{7}{2}x,\qquad g(L)=L^4-\tfrac{1}{2}L,$$
$$P_4(\mathcal{E},\mathcal{E})=g(3\alpha+\mathfrak{p})+\tfrac{7}{2}\alpha+\mathfrak{p}^4. \tag{F3.2}$$

Equality holds because the independent x-parameter fills the entire ideal $(7/2)\mathfrak{p}=\mathfrak{p}^4$. This is precisely why there can be no further higher-digit saving for these full inverse images.

Put $L_0=3\alpha$, $A_0=6L_0^2$, $B_0=4L_0$, and $\omega=P_4(\alpha,\alpha)$. The coefficient A_0 is a unit, and $g'(L_0)=4L_0^3-1/2\in 7\mathbf{Z}_7\subset\mathfrak{p}^3$, since $(3\alpha)^3=1 \bmod 7$. Taylor expansion, for $u\in\mathfrak{p}$, gives

$$g(L_0+u)-g(L_0)\equiv A_0u^2+B_0u^3 \pmod{\mathfrak{p}^4}.$$

Choose a uniformizer π and write $u=\pi b+\pi^2c \bmod \mathfrak{p}^3$, with digits $b,c\in\mathbf{F}_7$. The expression becomes

$$A_0b^2\pi^2+(2A_0bc+B_0b^3)\pi^3 \pmod{\mathfrak{p}^4}.$$

For $b=0$ there is one class. For $b\neq 0$, the leading square has three distinct nonzero values, and varying c fills all seven next digits because $2A_0b\neq 0$. Hence exactly $1+3\cdot 7=22$ classes occur modulo $\mathfrak{p}^4$. Multiples of seven in coefficients of π^2 and π^3 have valuations at least five and six, respectively, so digit reduction loses no mixed-characteristic carry.

Equivalently the actual infinite local support is the disjoint union

$$(\omega+\mathfrak{p}^4)\ \cup\ \bigcup_{i=1}^{3}(\omega+A_0\pi^2b_i^2+\mathfrak{p}^3), \tag{F3.3}$$

where b_i^2 are the three nonzero quadratic residues. Reducing this support gives four classes modulo $\mathfrak{p}^3$, and 22 fully filled classes modulo $\mathfrak{p}^4$. Each latter class has 7^{3m-4} descendants, proving (F3.1), $N_2=49+3\cdot 343=1078$, and $N_{m+1}=343N_m$ for $m\geq 2$. For $m\geq 2$, R_m has characteristic 7^m; replacing it with $\mathbf{F}_7[\pi]/(\pi^{3m})$ would be invalid.

F4 both input costs and the rational-modulus rate

PROVED count and FAILED positive-rate certificate. The raw output density for $m\geq 2$ is $22/7^4$. Each input fraction is $1/7$; charging both gives

$$\Psi_m=\frac{N_m/|R_m|}{(|E_m|/|R_m|)^2}=\begin{cases}4/7, & m=1,\\ 22/49, & m\geq 2.\end{cases} \tag{F4.1}$$

There are $d/3$ independent local cubic blocks, not d copies of this ramified quotient. The per-degree local gain therefore improves once from $\frac{1}{3}\log(7/4)$ to $\frac{1}{3}\log(49/22)$, then stabilizes. This is not optimization over arbitrary higher-level subsets.

The required ramified tower is a separate adaptation, not an assertion that the rational prime seven splits completely. The original cyclic cubic base includes the conductor prime seven and has one prime $\mathfrak{p}$ over it, with $7\mathcal{O}_F=\mathfrak{p}^3$. If its unramified pro-3 group has ranks $d_0\geq \ell-1$ and $r_0\leq d_0+C_0$, killing a Frobenius

at $\mathfrak{p}$ gives generator rank at least $d_0 - 1$ and relation rank at most $d_0 + C_0 + 1$. A Frattini element preserves generator rank; a non-Frattini element can be included in a generating basis and its generator eliminated in a free pro-3 presentation. For fixed sufficiently large ℓ,

$$d_0 + C_0 + 1 < (d_0 - 1)^2/4,$$

so Golod-Shafarevich keeps the quotient infinite. Only the prime of the base is forced to split in the relative layers; the original ramification index remains three. This is the construction proved in C3 §7, with its generator-elimination correction retained in the C2 precision addendum. The resulting tower has $7\mathcal{O}_{K_j} = \prod_{k=1}^{d/3} \mathfrak{P}_{j,k}^3$, constant $D \geq 7^{2/3}$, and identical local completions because the base prime splits in every relative layer. Set

$$\mathfrak{I}_j = \prod_k \mathfrak{P}_{j,k}, \quad N\mathfrak{I}_j = 7^{d/3}, \quad C_j = -1 + B_{\mathfrak{I}_j}(X).$$

The ideal lattice has covolume $7^{d/3} D^{d/2}$. Applying the half-box argument to this lattice gives

$$e^{ad} \leq |C_j| \leq e^{Bd}, \qquad a = \log X - \tfrac{1}{3}\log 7 - \tfrac{1}{2}\log D, \quad B = \log(2X + 1).$$

Fix $X = 2\,7^{1/3}\sqrt{D}$, giving $a = \log 2$. The bounded carrier is contained in the critical fibers; surjectivity onto them is neither asserted nor needed. A real embedding preserves all cardinalities. The directly proved C3 width on $[-1 - X, -1 + X]^2$ is

$$W = 81(X + 1)^4 + \tfrac{3}{2}X + \tfrac{27}{16}.$$

It comes from the convex maximum at a vertex and the one-variable minimum of $w^4 - w/2$ at $w = 1/2$. This proof works at $Q = 1$ and uses no missing C1 inequality. CRT and uniform separation give

$$|P_4(C_j, C_j)| \leq U_m = N_m^{d/3}(1 + W/7^m)^d.$$

With $C_* = (22/49)^{1/3}$, its signed rate is

$$\kappa_m = 2a - \frac{1}{d}\log U_m = \log \frac{X^2}{DC_*(W + 7^m)} \qquad (m \geq 2). \tag{F4.2}$$

For the diagonal polynomial $d(t) = P_4(t, t) = 81t^4 + 2t$, the average of the endpoint values minus the center value is

$$\frac{d(b + r) + d(b - r)}{2} - d(b) = 486b^2r^2 + 81r^4.$$

This quantity is at most the full continuous width, so that width is at least $81r^4$. Thus $(W + 7^m)/X^2 \geq 81X^2 + 7^m/X^2 \geq 18\,7^{m/2}$, and

$$\kappa_m \leq -\log(18DC_*7^{m/2}) < 0.$$

The same argument makes $\kappa_1 \leq -\log[18\sqrt{7}D(4/7)^{1/3}] < 0$. Level two can improve a negative level-one estimate; all levels above two strictly worsen (F4.2) at fixed radius. The ledger charges two windows, both ideal restrictions, discriminant, occupied classes, and worst-class population. Translation is in W; units, regulators, ratios, parameter fibers, denominator clearing and exceptions are inapplicable to this direct integral additive carrier. Mixed-characteristic carries were retained in F3.

For clarity, the construction ledger's nonzero contributions, in natural logarithms per field degree, are exactly

Entry	Contribution to the signed rate for $m \geq 2$
Two input windows	$+2\log X$
Critical input thinning, twice	$-\frac{2}{3}\log 7$
Field discriminant	$-\log D$
Exact output classes	$-\frac{1}{3}\log N_m$
Uniform population within each class	$-\log(1+W/7^m)$
Translation and coefficient height	Included in the proved width W
Real projection and parameter fibers	Zero loss: field embedding and translation are injective
Exceptional or uncovered outputs	None; every ordered pair is covered

Only the tower layer varies. The radius and the chosen rational precision are fixed beforehand; taking a degree-dependent precision does not convert the already saturated density into a new saving. If a new subcarrier lost input rate η, its pair budget would lose 2η. A replacement output rate c_{eff} would need $c_{\mathrm{eff}} < 2(a-\eta)$, with all additional branches and exceptions included. Then only exponents strictly below $[2(a-\eta)-c_{\mathrm{eff}}]/B$ are automatically certified when subexponential errors occur. None of F3's local proportions supplies this missing inequality.

F5 saturation, mixed depth, and the exact retirement scope

PROVED obstruction for the specified full-support/full-interval envelopes. The saturation ideal gives the sharper valid certificate

$$U_{\mathcal{J}^4} = 22^{d/3}(1+W/7^{4/3})^d, \quad \kappa_{\mathcal{J}^4} \leq -\log(18\,22^{1/3}D) < 0.$$

Using three shallow choices and one deep choice in every factor gives

$$U_{\mathrm{mix}} = \sum_{k=0}^{d/3} \binom{d/3}{k} 3^{d/3-k}(1+W/7^{1+k/d})^d.$$

All branches are included. Its all-shallow term already yields
$2a - d^{-1}\log U_{\mathrm{mix}} \leq -\log[18\sqrt{7}D(3/7)^{1/3}] < 0$.

More generally, let integral-ideal cosets, supported at seven, cover the **entire product** of the local supports (F3.3). In a common finite quotient, the product occupies fraction $(22/7^4)^{d/3}$; a coset of an ideal J_ℓ occupies fraction $1/NJ_\ell$. Subadditivity gives $\sum_\ell 1/NJ_\ell \geq (22/7^4)^{d/3}$. Consequently their uniform full-interval envelope satisfies

$$U_{\mathrm{cover}} = \sum_{\ell}(1+W/(NJ_\ell)^{1/d})^d \geq W^d(22/7^4)^{d/3}.$$

This also fails against actual nontrivial equal-radius critical-coset population. Ideal separation gives $|C| \leq (1+2r/7^{1/3})^d$; $|C|>1$ forces $z = 2r/7^{1/3} \geq 1$. Together with $W \geq 81r^4$,

$$U_{\mathrm{cover}}/|C|^2 \geq [81\,22^{1/3}z^4/(16(1+z)^2)]^d \geq [81\,22^{1/3}/64]^d > 3.54^d.$$

These are lower bounds on numerical envelopes, never on the actual image. The hypotheses include entire local product support, full continuous intervals on every branch, uniform ideal-norm packing, and the stated input population normalization. A proper subset of globally realized residues, smaller branch-dependent real widths, correlated factors, arbitrary deeper input subsets, or another carrier lies outside this retirement. The local theorem alone supplies none of these improvements. The illustrative floor $D = 7^{2/3}$ is not a claim that the constructed tower attains that value.

F6 degree four can succeed, without solving T02

PROVED elementary consequences. A quadratic polynomial with compressed image remains compressed after composition with any fixed univariate polynomial. For the reconstructed rank-one quadratic baseline $f(x, y) = x + y + (x - y)^2$, the degree-four polynomial $f(x, y)^2$ satisfies $|f(A, A)^2| \leq |f(A, A)|$ on the same witnesses. Likewise pure $(x - y)^4$ has exactly n values on $A = \{0, \ldots, n - 1\}$. More generally, fixed integer linear forms raised to a power have $O(n)$ image on integer progressions. These are composition/additive-form examples, not transverse quartic mixtures such as T02. Degree alone is not the boundary.

The broader C1 translated/anisotropic interval theorem and independent coordinate-thinning inequality are **PROVED as reported in the controlling record**, but their complete original main proof was not received in the curated local supplement. The recovery addendum identifies a larger package whose original files were not part of the received intake. This chapter does not independently verify that unavailable proof or use it as a dependency of F2-F5; the symmetric obstruction and C4/C3 arguments above are inspectable.

STRUCTURE PROVED / ENTIRE SUPPORT OPEN

4 Norm-one directions and the missing cubic bound

N1 - PROVED relative to the source arithmetic supply. In the CM field $K = L(i)$, complex conjugation c exchanges the td selected prime pairs. Binary choices give ideals $\mathfrak{a}_\epsilon$. A largest ideal-class fiber, anchored at η, permits $(\alpha_\epsilon) = \mathfrak{a}_\epsilon \mathfrak{a}_\eta^{-1}$ and $u_\epsilon = \alpha_\epsilon / c(\alpha_\epsilon)$. Then

$$uc(u) = 1, \quad v_{\mathfrak{p}_s}(u) = 2(\epsilon_s - \eta_s), \quad Q_0^2 u \in \mathcal{O}_K, \qquad Q_0 = \prod_b q_b,$$

$$e^{\gamma d} \le |U| \le 2^{td}, \qquad \gamma = t \log 2 - \log H > 0. \tag{N.1}$$

Sources: checkpoint §§C.2/H.2, export §4.5 and formal source Proposition 2.2; the supplied bound is $h(K) \le H^d$. There is no new arithmetic input or assertion of a prescribed numerical γ.

Proof of the deductions. Ideal-class pigeonholing retains at least $2^{td}/H^d$ choices. Distinct binary vectors give distinct displayed valuations, proving distinctness. The valuations bound the poles by Q_0^2, and CM conjugation gives modulus one in every complex embedding. For $t(u) = u + u^{-1}$, trace equality implies $v = u$ or $v = u^{-1}$. The latter requires $\epsilon + \epsilon' = 2\eta$, which forces both vectors to equal η. Hence t is injective on this U, and $T = t(U) \subset L$ has $|T| = |U|$ with conjugates in $[-2, 2]$.

The export §4.5 additionally records $|UU|, |U/U| \le 4 \cdot 3^{td}$ from the source's ideal labels. This supporting arithmetic bound is retained as reported, with its original proof input identified; no new class-number or unit-theorem audit is claimed. It explains the attraction of product and ratio parameters: each has three valuation possibilities where the inputs have four ordered binary choices. N2 below shows why counting them separately is insufficient. Their joint valuation information recovers both ordered input vectors.

N2 - PROVED joint identities; no parameter saving. For all ordered $(u, v) \in U^2$, let $p = t(uv)$, $s = t(u/v)$, $D = t(u) - t(v)$, $E = t(u) + t(v)$. Expansion gives

$$D^2 = (p - 2)(s - 2), \qquad E^2 = (p + 2)(s + 2), \qquad P_3 = D^3 + E. \tag{N.2}$$

The joint map $(u, v) \mapsto (p, s)$ has fibers at most eight. Source: §H.2; dependency N1.

Proof. Each trace determines its nonzero argument up to inversion, so there are at most four choices of $(uv, u/v)$. Each determines $u^2 = (uv)(u/v)$, with at most two roots, then determines v. Thus the joint parameter image has at least $|U|^2/8$ states. The identities retain both parameters and signed roots; counting this joint support alone gives only a constant-factor reduction.

N3 - PROVED fixed-label fiber bound; full cubic fibers remain OPEN. Fix either $uv = \xi$ or $u/v = \xi$, and a cubic output z. There are at most six ordered pairs satisfying both constraints. Source: §H.2; dependencies N1-N2.

Proof. In both cases $t(v) = \xi/u + u/\xi$. Put $a = 1 - \xi^{-1}, b = 1 - \xi, c = 1 + \xi^{-1}, e = 1 + \xi$. Then $D = au + b/u$, $E = cu + e/u$. Multiplying $D^3 + E = z$ by u^3 gives

$$a^3 u^6 + (3a^2 b + c) u^4 - z u^3 + (3ab^2 + e) u^2 + b^3 = 0.$$

If $\xi \neq 1$, its leading coefficient is nonzero. For $\xi = 1$, the polynomial is $2u^4 - zu^3 + 2u^2$, also nonzero. Its degree is at most six; the fixed label determines v from u. Consequently

$$|P_3(T,T)|\,|UU| \geq |U|^2/6, \qquad |P_3(T,T)|\,|U/U| \geq |U|^2/6.$$

These are not quadratic lower bounds for the actual image. A full cubic fiber of size r must bridge at least $r/6$ products and $r/6$ ratios. Separate small product or ratio sets do not prove the required overlap between different labels.

N4 - PROVED container; FAILED certificate. Since $Q_0^2 T \subset \mathcal{O}_L$, $Q_0^6 P_3(T,T) \subset \mathcal{O}_L$. The bound $|\sigma(P_3)| \leq 4^3 + 4 = 68$ and integral-norm packing give

$$|P_3(T,T)| \leq (136 Q_0^6 + 1)^d.$$

For the source primes $q_b \geq 5$, this expression exceeds $Q_0^{6d} \geq 5^{6td} > 2^{2td} \geq |T|^2$. Source: §H.2; dependencies N1 and the packing argument of C5.

The pole is genuine at a coordinate where labels vary: an active trace has valuation -2, whereas the anchored inactive trace is integral. Their difference-cube has valuation -6, and their sum has valuation -2; cancellation is impossible. Norm one therefore does not uniformly remove cubic denominator growth. This proves failure of the displayed full-denominator container, not of every finer count.

For real coordinates $r_u = t(u)/2$, (N.2) gives $8P_3(r_u, r_v) = D^3 + 4E$. The useful planar identity

$$(r_u - r_v)^2 + (s_u - s_v)^2 = 2 - t(u/v), \qquad s_u = (u - u^{-1})/(2i),$$

controls squared distances, not the scalar cubic. Also $P_3(a, a-h) = h^3 + 2a - h$ is injective in the anchor a. Selected unit-distance translations leave all other ordered pairs uncontrolled; no exact same-set cubic transport is supplied.

The missing entire support statement

N5 - PROVED implications; OPEN source-specific premises. Applied to the actual trace family, an entire-image estimate

$$\log |P_3(T_d, T_d)| \leq bd + o(d), \qquad b < 2\gamma,$$

would certify every fixed $0 < c < (2\gamma - b)/(t \log 2)$, by (N.1). This is the charged rate gate in the 11 October `A_transfer.md` §A3: divide by the squared input lower count and use $\log |T_d| \leq (t \log 2) d$. Any thinning or parameter-to-scalar loss reduces the input rate twice in the pair budget. No such b is established for this family.

A relative formulation is equally explicit. For fixed $C > 0$, $m \geq 0$ and $\kappa > 0$, the source-carrier-specific premise

$$|P_3(T_d, T_d)| \leq C d^m e^{-\kappa d} n_d^2, \qquad n_d = |T_d|, \tag{N.3}$$

would give every fixed $0 < c < \kappa/(t \log 2)$ eventually. Indeed $\log n_d \leq (t \log 2) d$, and the strict inequality absorbs $\log C + m \log d$. Equation (N.3) is **OPEN** for this carrier, not a simpler lemma already supplied by the unit-distance paper.

The exact identities N2 also give a signed-radical presentation:

$$P_3(t(u),t(v)) = \sigma\{(p-2)(s-2)\}^{3/2} + \tau\sqrt{(p+2)(s+2)}, \qquad \sigma,\tau \in \{-1,1\}. \tag{N.4}$$

The radicands are nonnegative because they are D^2, E^2; the signs come from the actual D, E, with coinciding choices at zero. Only states attained by actual ordered pairs are counted, not an independent product of product traces and ratio traces. A bound such as (N.3) for this entire attained radical support would suffice. N2's bounded joint fibers and N3's fixed-label root bounds do not prove it: the required coincidences between different attained states remain OPEN.

For the existing sufficient fiber route, let $n = |A|$ and $r(z) = \#\{(a,b) \in A^2 : P_3(a,b) = z\}$. The exact identity

$$|P_3(A,A)| = \sum_{(a,b)\in A^2} \frac{1}{r(P_3(a,b))}$$

shows that, if all but $O(n^{2-\kappa})$ pairs have actual full fiber at least n^{ϵ}, then the image is $O(n^{2-\kappa} + n^{2-\epsilon})$. This includes isolated pairs and all cross-pairs. Source: latest `C_entropy.md` §C5, equations (C.8)-(C.9). Large collision energy or a few selected rich fibers does not supply this tail.

For the trace carrier, N3 makes the needed overlap especially explicit. A rich actual cubic fiber must contain a growing number of distinct products and ratios, despite the bounded number of pairs contributed by each one. A theorem about one product label, one direction, or a selected principal ideal-class subfiber leaves this obligation unresolved. Any excluded ordered pairs must be counted as exceptions in the tail; a fixed positive fraction of excluded pairs contributes order n^2 possible outputs and therefore gives no positive exponent by this argument. This is a necessary bookkeeping condition for this certificate, not a new obstruction against an unidentified carrier.

The 11 October record retains **EG-O1 / NC-O1** as this OPEN actual all-pair fiber-tail obligation; it closes only the finite CP-O1 attachment. The separate 10 September norm-trace checkpoint concerns $P_0 = x^2 + xy + y^2 + x$, not P_3. Its quadratic norm identities, conditional same-set transfer, and norm-action tests supply no cubic support bound. No additional candidate or research assignment is created by this report.

EXTERNAL ARITHMETIC INPUT

5 What the original arithmetic construction supplies

A1 Source identity and audit boundary

EXTERNAL RESULT / inspected source system. The controlling original is OpenAI, *Planar Point Sets with Many Unit Distances*. Its Theorem 1.1 concerns planar point sets with a superlinear number of unit distances, rather than a small scalar polynomial image. The supplied copy is `local-only/migration/research_supplement/shared/03-unit-distance-proof.pdf`; its verified SHA-256 is `62b5de8cacdfbc0f13ae1940aa81659cd233e6945b936612fbeab0ab8ab3d383`. The official PDF URL was also accessed. No byte-identity assertion between that response and the local copy is made. The PDF has no printed date or version; its creation metadata is not used as mathematical chronology.

This audit read the complete relevant printed pages 6-16. Pages 6-9 and 12-14 were additionally rendered and visually inspected. Bibliographic statements below are the original paper's identified imports, not fresh verification of the cited books or papers. In particular, an original-paper proposition label is distinguished from a theorem number in the underlying reference.

All A1-A8 records belong to this file and version date. Their evidence type is source inspection, with elementary consequences explicitly identified where used. Dependencies and retained gaps appear with each record. No external novelty claim is made.

A2 Cyclic cubic base field and its discriminant

EXTERNAL RESULT / original Proposition 3.2, pp. 10-11; Appendix A.11, p. 16. Choose distinct rational primes $r_i \equiv 1 \pmod 3$, for $1 \le i \le \ell$. Let L_i be the cyclic cubic subfield of $\mathbf{Q}(\zeta_{r_i})$, choose cubic characters χ_i of conductor r_i, and let F be the cyclic cubic field cut out by $\prod_i \chi_i$ inside $M = L_1 \cdots L_\ell$. With $R_\ell = \prod_i r_i$, the imported cyclotomic facts and conductor-discriminant formula give

$$\mathrm{Gal}(M/\mathbf{Q}) \cong (C_3)^\ell, \quad \mathrm{Gal}(M/F) \cong (C_3)^{\ell-1}, \quad |D_F| = R_\ell^2, \quad |D_M| = R_\ell^{2\cdot 3^{\ell-1}}.$$

Both fields are totally real. The equality $|D_M| = |D_F|^{[M:F]}$, together with the relative discriminant formula, makes M/F everywhere unramified. Thus it supplies an elementary abelian unramified 3-extension of rank $\ell - 1$. The source cites Washington, *Introduction to Cyclotomic Fields*, Chapter 3, especially Theorem 3.11, and Neukirch, Chapter VI. These references are retained as source locators, rather than reassigned theorem labels.

For the first ℓ primes $r_i \equiv 1 \pmod 3$, equation (6), p. 13, uses the prime number theorem in arithmetic progressions to obtain

$$\log \mathrm{rd}(F) = \frac{2}{3} \sum_i \log r_i = O(\ell \log \ell).$$

Dependency: A1 and the identified cyclotomic/discriminant imports. The root-discriminant parameter is attached to these auxiliary primes. It is not the product of the subsequently chosen completely split primes.

A3 Relations, generators and survival of the tower

EXTERNAL RESULTS / original Propositions 3.3-3.5, p. 11; Appendix A.8-A.10, p. 16. The sufficient forms are:

1. For a finitely generated pro-p group G, write $\Phi(G) = \overline{G^p[G,G]}$ for its closed Frattini subgroup. Then $d(G) = \dim_{\mathbf{F}_p} G/\Phi(G)$. If k elements of $\Phi(G)$ are killed by their closed normal closure N, the quotient retains its generator rank and adds at most k relations:

$$d(G/N) = d(G), \qquad r(G/N) \le r(G) + k.$$

   Relation rank is used for the finitely presented groups in the arithmetic application. The original cites Ribes-Zalesskii §2.8, Koch Theorem 4.10, and Dixon-du Sautoy-Mann-Segal Proposition 1.9(ii).

2. For F totally real cubic, $\zeta_3 \notin F$, and $G = \mathrm{Gal}(F^{\mathrm{ur},3}/F)$, Shafarevich's estimate in the form used here is

$$r(G) \le d(G) + C_0$$

   for an absolute constant C_0. Here $F^{\mathrm{ur},3}$ means the maximal everywhere-unramified pro-3 extension, including infinite places. The source cites Shafarevich's 1963/1966 papers and Neukirch-Schmidt-Wingberg, Chapter X §10; it supplies no more precise underlying theorem number for this specialization.

3. A finite nontrivial pro-p group with generator rank d and relation rank r satisfies $r > d^2/4$. Consequently, a nontrivial finitely generated pro-p group with $r \le d^2/4$ is infinite. This is original Proposition 3.4, the Golod-Shafarevich criterion, citing Golod-Shafarevich 1964/1965 and Koch Chapter 11.

Dependency: A1; A2 supplies $d(G) \ge \ell - 1$. These are imported general results in the source system, not independent proofs in this report.

A4 Finite Chebotarev choice and what it means

EXTERNAL RESULT / original Proposition 3.6, p. 11; Appendix A.12, p. 16. In a prescribed finite Galois extension of $\mathbf{Q}$, infinitely many rational primes split completely, even after a fixed finite exclusion. Apply this to the normal closure of $E(i)$, where E/F corresponds to the finite Frattini quotient $G/\Phi(G)$. For any finite requested number t, this selects distinct $q_1, \dots, q_t$ with

$$q_b \equiv 1 \pmod 4, \qquad q_b \text{ completely split in } F, \qquad \mathrm{Frob}_v \in \Phi(G) \quad (v \mid q_b).$$

The source locators are Neukirch, Chapter VII §13, and Tschebotareff's 1926 paper. The complete-splitting case is all the tower argument needs; no effective bound on the sizes of the q_b is asserted here.

Dependency: A1-A3. These primes are **chosen after** F and E have been fixed. The result does not guarantee an arbitrary preassigned list of small primes. Splitting at this one finite level does not alone imply splitting in every higher layer: the Frobenius-killing quotient in A5 provides that step. The stated congruence is $1 \bmod 4$, with no stated $1 \bmod 3$ requirement. Selecting primes through the finite field $E(i, \zeta_3)$ instead would impose $1 \bmod 12$ by the same Chebotarev principle; that is an adaptation of the argument, not the proposition's literal statement.

A5 The assembled original tower and parameter order

EXTERNAL RESULT / original Proposition 3.8, pp. 12-14; Remark 3.9, p. 14. For every sufficiently large fixed integer ℓ, put

$$t = \lfloor (\ell - 1)^2/100 \rfloor.$$

There are a totally real cyclic cubic base field F, the fixed selected $q_1, \dots, q_t$, and a tower $F = F_0 \subset F_1 \subset \cdots$ satisfying: each F_j/F is finite Galois, everywhere unramified, with a 3-group Galois group; $f_j = [F_j : \mathbf{Q}] \to \infty$; every F_j is totally real; and

$$\mathrm{rd}(F_j) = \mathrm{rd}(F), \qquad q_b \text{ completely split in every } F_j.$$

The proof outline explains the quantitative compatibility. A2 gives $d = d(G) \geq \ell - 1$. Each selected rational q_b has exactly three primes above it in F. Killing their Frobenius representatives therefore adds at most $3t$ relations. By A3 and A4 the quotient $\overline{G}$ has

$$d(\overline{G}) = d, \qquad r(\overline{G}) \leq d + C_0 + 3t \leq d + C_0 + 3d^2/100 < d^2/4$$

for sufficiently large ℓ. It remains infinite by Golod-Shafarevich. Finite layers of this quotient have trivial selected Frobenius, which gives complete splitting at every layer. Unramified extensions preserve root discriminant. An odd-order extension cannot turn a real place complex.

For $K_j = F_j(i)$, adjoining i gives

$$\mathrm{rd}(K_j) \leq 2\,\mathrm{rd}(F).$$

EXTERNAL RESULT / original Proposition 3.7, p. 12; Appendix A.13, p. 16. The sufficient class-number estimate for any number field K is

$$h(K) \leq \max\{2, \mathrm{rd}(K)\}^{C_{\mathrm{class}}[K:\mathbf{Q}]}$$

with an absolute constant C_{class}. It comes from Minkowski's ideal-class bound and an ideal-counting/divisor-function estimate, with Neukirch Chapter I §5 and Lang Chapter V as source locators. Consequently $h(K_j) \leq H_\ell^{f_j}$, with $H_\ell = (2\,\mathrm{rd}(F))^{2C_{\mathrm{class}}}$ and $\log H_\ell = O(\ell \log \ell)$. Thus

$$\gamma = t \log 2 - \log H_\ell > 0$$

eventually: t grows quadratically in ℓ, while the logarithmic discriminant and class-number costs grow as $O(\ell \log \ell)$.

Dependencies: A2-A4 and the identified class-number estimate. This is an existence theorem for arbitrarily large **fixed finite lists**, with the tower depending on the construction parameters. It makes no original-paper claim of infinitely many common split primes in one tower. Remark 3.9 fixes ℓ, then the q_b and $Q = \prod_b q_b$, then the window radius; only afterward does j vary. Large selected primes affect Q and subsequent packing/exponent constants, even though they do not raise $\mathrm{rd}(F)$ in this construction.

A6 What the original norm one construction supplies

EXTERNAL RESULT / original Definition 2.1 and Proposition 2.2, pp. 6-7. For an admissible datum L, $K = L(i)$, fixed completely split $q_b \equiv 1 \pmod 4$, $f = [L : \mathbf{Q}]$, and $h(K) \leq H^f$, set $Q = \prod_b q_b$. Each rational prime gives f conjugate prime-ideal pairs $\{\mathfrak{P}_s, c\mathfrak{P}_s\}$ in K, so there are tf pairs. The 2^{tf} binary choices of one member of each pair occupy at most $h(K)$ ideal classes. Ratios within one largest class fibre are principal. The source constructs distinct elements $u_\varepsilon = \alpha_\varepsilon / c(\alpha_\varepsilon)$, with

$$uc(u) = 1, \quad |\sigma(u)| = 1 \text{ for every complex embedding}, \quad U \subset Q^{-2}\mathcal{O}_K, \quad |U| \geq e^{\gamma f}.$$

Distinctness and denominators follow from valuations $v_{\mathfrak{P}_s}(u_\varepsilon) = 2(\varepsilon_s - \eta_s)$ and their negatives at conjugate primes. These are norm-one elements with controlled poles, not a claim that every element is an algebraic-integer unit. Dependencies: A5 for a growing sequence and $\gamma > 0$. In this section the original notation $D = Q^2$ denotes a **denominator**, not root discriminant.

A7 The selected-translation theorem and its packing cost

EXTERNAL RESULT / original Theorem 2.3 and Lemmas 2.4-2.6, pp. 7-9. Embed $D^{-1}\mathcal{O}_K$ in $\mathbf{C}^f$, using one complex embedding over each real embedding of L. For the resulting lattice Λ, let $X_a = (a+\Lambda)\cap B_R$, where B_R is a product of radius-R discs. Write $N_a = |X_a|$ and count only selected translations:

$$E_a = \#\{(x,x') \in X_a^2 : x' - x \in U\}.$$

If ρ_R is the overlap-area ratio for two such discs with centres one unit apart, averaging gives a coset with $E_a \geq |U|\rho_R^f N_a$. Fix $R > 1/2$ so that $\log \rho_R > -\gamma/2$. Then

$$E_a \geq e^{\gamma f/2} N_a, \qquad e^{\gamma f/2} \leq N_a \leq (1+2RD)^{2f} \leq e^{Bf}, \quad B = 2\log(4RD).$$

Projection onto one **complex** coordinate is injective on the lattice coset, because its differences are field elements. It gives a planar set of the same cardinality and at least $\frac{1}{2}e^{\gamma f/2}N_a$ unit distances. Theorem 2.3 obtains a fixed saving parameter $\delta = \gamma/(4B) > 0$ for that unit-distance assertion. Dependencies: A6 and original averaging, field-embedding injectivity and norm-based packing. This is selected-edge abundance; it controls neither the complement of those edges nor every scalar polynomial output on a complete Cartesian square.

A8 Infinite splitting, scalarization and retained obligation

Separate stronger EXTERNAL RESULT / optional. The human-authored *Remarks on the Disproof of the Unit Distance Conjecture*, Proposition 2.3, printed p. 6, states that one bounded-root-discriminant totally real tower can have infinitely many completely split rational primes $q \equiv 1 \pmod 4$. Its recursive Chebotarev/cutting argument uses $F(i)$ whenever a split prime of a finite field F is selected. The remarks cite Hajir-Maire-Ramakrishna, arXiv Theorem 2.8 / published Theorem 4. That underlying paper was not independently checked here. The remarks PDF was read through the web; no local byte hash or printed version date is asserted. This stronger input must not be silently attributed to A5. Pohoata's 2026 polynomial-image paper is a separate secondary comparison in the wider report. Its polynomial theorem and constants are not conclusions of the original unit-distance Theorem 1.1.

Elementary consequences inspected here; full scalar support OPEN. There is no field embedding $L(i) \to \mathbf{R}$. Taking real parts of a full complex window loses injectivity: its field-difference kernel is iL. For U alone, however, put $r(u) = (u + c(u))/2$ and $s(u) = (u - c(u))/(2i)$. They belong to $(2D)^{-1}\mathcal{O}_L$, all their real conjugates have modulus at most one, and $r(u)^2 + s(u)^2 = 1$. A prescribed real part has at most two norm-one preimages, so $|r(U)| \geq |U|/2$. The circle relation concerns matched coordinates of one u, not every pair in $r(U)^2$; multiplication and the original unit-distance geometry are not preserved by real-part projection.

For T01 the missing requirement remains a same-set real carrier with lower input rate $a > 0$ and an entire cubic-image upper rate $c < 2a$, with every projection fibre, denominator, exception and uncovered pair charged. An exponential input loss $e^{\eta f}$ changes the pair budget to $2(a-\eta)$. A6-A7 supply no such full-support theorem. Even on a fixed difference slice, $P_3(a, a+d) = -d^3 + 2a + d$ is injective in a. Cross-slice collisions require additional control. The original arithmetic tower survives; the requested cubic support certificate remains **OPEN**.

COUNTING FOUNDATIONS / G1-G8

Appendix A Support and rate lemmas

The field degree is denoted by d, the root-discriminant bound by D, a rational residue modulus by m, and the radius of a centered input box by X. These are different quantities. In a tower, all parameters other than its layer and degree are fixed first.

Lemma G1 Integral norm separation

PROVED; elementary argument inspected. Let K be totally real of degree d. If distinct $z, z' \in \mathcal{O}_K$ are congruent modulo $m\mathcal{O}_K$, some real embedding satisfies

$$|\sigma_i(z - z')| \geq m.$$

Proof. Write $z - z' = m\alpha$, where $0 \neq \alpha \in \mathcal{O}_K$. Its norm is a nonzero integer, so $\prod_i |\sigma_i(\alpha)| \geq 1$. At least one factor is at least one. This proves separation in the maximum norm over all embeddings. It gives no degree-independent separation in one chosen real embedding. □

Lemma G2 Input counts in a centered box

PROVED; complete elementary proof. Define

$$B_K(X) = \{\alpha \in \mathcal{O}_K : |\sigma_i(\alpha)| \leq X \text{ for every } i\}.$$

If $\mathrm{rd}(K) \leq D$, then for $X \geq 1$,

$$\boxed{(X/\sqrt{D})^d \leq |B_K(X)| \leq (2X+1)^d.}$$

Proof. The Minkowski lattice has covolume $\sqrt{|\Delta_K|}$. Blichfeldt averaging on a cube of side X produces a translate containing at least $X^d/\sqrt{|\Delta_K|}$ lattice points. Subtract one of these points from every point in that translate. The resulting distinct lattice points lie in the centered cube of radius X. This proves the lower count. Lemma G1 with $m = 1$ permits disjoint half-open cubes of side one around all input points; their union is contained in a cube of side $2X + 1$. Comparing volumes proves the upper count. □

Thus $X > \sqrt{D}$ ensures unbounded input size when $d \to \infty$. The lower count uses the discriminant; an archimedean upper bound alone does not establish enough inputs.

Lemma G3 Uniform output population in every residue class

PROVED; complete elementary proof. Suppose all conjugates of the outputs lie in intervals of lengths $W_1, \ldots, W_d$, and let $\Omega \subset \mathcal{O}_K/m\mathcal{O}_K$ be their allowed residue classes. Then

$$\#\{z \in \mathcal{O}_K : z \bmod m \in \Omega,\ \sigma_i(z) \in I_i \text{ for every } i\} \leq |\Omega| \prod_{i=1}^{d} (1 + W_i/m).$$

In particular, if $|\sigma_i(z)| \leq H$, the bound is $|\Omega|(1 + 2H/m)^d$.

Proof. By G1 the points in any one residue class are m-separated in maximum norm. Center disjoint half-open cubes of side m at them. Their union lies in the product of intervals enlarged by $m/2$ at each end, with volume $\prod_i (W_i + m)$. Divide by the volume m^d of one cube, then sum over all allowed classes. □

This is a worst-case estimate per residue class. It assumes neither equidistribution of the input box nor statistical independence of its reductions.

Lemma G4 Complete splitting and vertical amplification

PROVED from CRT and the definition of complete splitting. Suppose each distinct rational prime p_i splits completely in K. Then for fixed positive integers r_i,

$$\mathcal{O}_K/\Big(\prod_i p_i^{r_i}\Big)\mathcal{O}_K \cong \prod_i (\mathbb{Z}/p_i^{r_i}\mathbb{Z})^d.$$

For $r_i = 1$ this is the product of d copies of $\mathbb{F}_{p_i}$ for each prime. For higher powers, complete splitting means that the local fields are $\mathbb{Q}_{p_i}$, so their integral quotients are $\mathbb{Z}/p_i^{r_i}\mathbb{Z}$. CRT combines the factors.

If every output belongs to a specified subset of density η_i in every coordinate above p_i, then, with $m = \prod_i p_i^{r_i}$ and $\eta = \prod_i \eta_i$,

$$|\Omega| \le (\eta m)^d.$$

The quotient coordinates are independent algebraic factors. The actual bounded carrier may occupy a correlated subset of them, which can only reduce this allowed output set. For other splitting types, one must count the actual residue degrees, ramification and number of factors; d degree-one copies cannot be assumed.

Lemma G5 The fully charged box criterion

PROVED implication; its antecedents must be established for the actual carrier. Let $F \in \mathbb{Z}[x, y]$, use $B = B_K(X)$, and suppose every output belongs to at most $(\eta m)^d$ residue classes and has all conjugates bounded by $H(X)$. Then

$$|F(B, B)| \le [\eta(2H(X) + m)]^d \le \Theta(X)^d |B|^2,$$

$$\boxed{\Theta(X) = D\eta \frac{2H(X) + m}{X^2}.}$$

Proof. Apply G3 and G4, then square the lower bound in G2. □

If $\Theta < 1$, $X > \sqrt{D}$, and a proved same-set real transport preserves input and image cardinalities, G2 gives the exact exponent

$$c = \frac{-\log \Theta}{\log(2X + 1)} > 0.$$

Indeed $|B| \le (2X + 1)^d$ implies $\Theta^d \le |B|^{-c}$. If this numerical certificate has $\Theta \ge 1$, that certificate fails; it does not lower-bound the occupied image. A local defect $\eta < 1$, complete splitting and bounded root discriminant do not alone imply $\Theta < 1$.

Lemma G6 General rates and the cost of input losses

PROVED implication; complete proof. Suppose a family of actual real scalar sets satisfies

$$\log |A_d| \ge ad - o(d), \quad \log |A_d| \le Bd + o(d), \quad \log |P(A_d, A_d)| \le bd + o(d),$$

where $a > 0$, $B > 0$, and $\delta = 2a - b > 0$. Every fixed $0 < c < \delta/B$ is eventually certified.

Proof. The image divided by $|A_d|^2$ is at most $e^{-\delta d+o(d)}$. Since $\log|A_d| \le Bd+o(d)$, a strict choice $cB<\delta$ absorbs all error terms and any fixed or polynomial prefactors. □

If thinning, projection or parameter fibers lose $e^{\lambda d+o(d)}$ inputs, replace a by $a-\lambda$. The gap decreases by 2λ. An unknown loss is not zero. Boundary exponents require additional error control; they do not follow from this asymptotic lemma. G5 has an exact finite bound and therefore has stronger boundary control in that particular situation.

Lemma G7 Common scaling and finite local images

PROVED; elementary identities. For a nonzero integer Q, if $P=\sum_{j=0}^{k}P_j$, with P_j homogeneous of degree j, the unique polynomial of degree at most k satisfying $F_Q(Qx,Qy)=Q^kP(x,y)$ is

$$F_Q=\sum_{j=0}^{k}Q^{k-j}P_j.$$

This follows by comparing homogeneous pieces. Its reduction at $p\mid Q$ is the leading homogeneous part P_k. For a real embedding and an algebraic carrier B, the correct normalized set is $A=Q^{-1}\sigma(B)$; then the image cardinalities agree. Scaling alone gives no additional compression.

For a nonzero linear form L over $\mathbb{F}_p$,

$$|L(\mathbb{F}_p,\mathbb{F}_p)^k|=1+\frac{p-1}{\gcd(k,p-1)}.$$

Proof. L is surjective. The cyclic group $\mathbb{F}_p^\times$ has a power-map kernel of size $\gcd(k,p-1)$, and zero supplies one additional value. □

For an odd prime $p\nmid k$, the number of k-th powers modulo p^r is

$$1+\frac{1}{g}\sum_{h=0}^{\lfloor (r-1)/k\rfloor}\varphi(p^{r-kh}),\qquad g=\gcd(k,p-1).$$

Every nonzero power has valuation $kh<r$; its unit part is a k-th power in the cyclic unit group at the remaining precision, with kernel g. Different valuations give disjoint values. The density tends to

$$\frac{p-1}{gp(1-p^{-k})}>0.$$

Consequently greater precision at one good fixed prime does not supply a density tending to zero for this full power-residue support.

The undeformed T01 polynomial has full image over every odd finite field: in coordinates $h=x-y$, $s=x+y$, it is h^3+s. For T02, the change $(x,y)\mapsto(x+2y,3x-y)$ has determinant -7, so the undeformed polynomial has full field image at $p\neq 7$. The residue permutation in F7 also covers the field at seven. The useful power-residue defect is therefore a property of the removable deformation, or of a more refined restricted-input quotient; it is not a sparsity assertion about the original unrestricted field image.

Lemma G8 Restricted inputs and exact output fibers

PROVED; elementary bookkeeping. For a finite ring R, a same-set input $E \subset R$, and its actual polynomial image, the local factor relative to the pair budget is

$$\kappa(R, E, F) = \frac{|R|\,|F(E,E)|}{|E|^2}.$$

This is output density divided by the square of input density. Product factors multiply only when the construction and all local restrictions are proved to have that product structure. A finite product ring does not admit an injective unital ring map into $\mathbb{R}$ when it has nontrivial zero divisors; a quotient calculation therefore still requires a faithful arithmetic realization and a full rate ledger.

For any finite real A, set $r(z) = |\{(a,b) \in A^2 : P(a,b) = z\}|$. Then

$$|P(A,A)| = \sum_{(a,b)\in A^2} \frac{1}{r(P(a,b))}.$$

Each nonempty fiber contributes one. If all but $O(n^{2-\eta})$ ordered pairs have actual fiber at least n^{ε}, then

$$|P(A,A)| \le O(n^{2-\eta}) + n^{2-\varepsilon}.$$

Large collision energy alone gives the opposite Cauchy-Schwarz inequality, $|P(A,A)| \ge n^4 / \sum_z r(z)^2$. Small Shannon entropy and coarse covering numbers also do not upper-bound exact support without an exact population bound for each cell. These are support requirements, not extra saving factors.

Sources for G1-G8. Corrected C0 §§4-6; the cubic checkpoint §§B, E-F; the 2026-10-11 transfer note §§A2-A3 and entropy note §C5. The statements and elementary proofs above are inspected consolidations. Their source identities and audit scopes appear in the claim ledger.

LOCAL CLASSIFICATION / Q3-Q5

Appendix B Full rank quadratics and finite intervals

Lemma Q5 The finite interval comparison

PROVED elementary interval bound; a separate comparison with Pohoata Proposition 2.1 and Corollary 2.2. For $[N] = \{1, \dots, N\}$, fixed distinct odd primes with product Q, and their square density ρ,

$$|f_Q([N], [N])| \leq \rho(2QN + N^2 + Q).$$

Proof. The integer outputs lie in an interval of length at most $2QN + N^2$ and occupy at most ρQ classes modulo Q. Each class supplies at most the interval length divided by Q, plus one. □

The prime number theorem permits $t = (1 + o(1)) \log N / \log\log N$ primes in $[\log N, 2\log N]$, with $Q \leq N$. The product of their factors $1 + 1/p$ is $e^{o(1)}$, so the bound becomes

$$N^2 \exp\left[-(\log 2 - o(1)) \frac{\log N}{\log\log N}\right].$$

Exact rescaling to $A = [N]/Q$ keeps f fixed even when Q varies. The saving furnished by this particular interval comparison has exponent tending to zero. It does not prove a fixed positive exponent. Thus a varying auxiliary modulus is not intrinsically invalid when an exact transport removes it; the missing issue is a uniform positive rate. The tower replaces growing rational primes with growing numbers of prime-ideal coordinates while keeping the remaining costs fixed.

Lemma Q3 Full rank binary quadratics at good primes

PROVED; complete finite-field argument. Over $\mathbb{F}_p$ with odd p, a homogeneous binary quadratic form has image $\{0\}$ at rank zero, a set of $(p+1)/2$ elements at rank one, and the entire field at rank two.

Proof. Rank one is a nonzero scalar times the square of a surjective linear form. An isotropic rank-two form is equivalent to λuv, which represents every value. An anisotropic rank-two form is equivalent to $\lambda(u^2 - av^2)$, with a nonsquare. This is a scalar multiple of the norm $\mathbb{F}_{p^2} \to \mathbb{F}_p$. On the cyclic multiplicative group, the norm $z \mapsto z^{p+1}$ has image of size $p - 1$; zero is also represented. □

For an integral nondegenerate quadratic form, only finitely many primes divide its determinant, discriminant or content. At the other odd primes, adding linear and constant terms preserves surjectivity by completing the square and translating the inputs and output. The common-scaling deformation in G7 therefore has no unrestricted first-level density defect at good primes.

In particular $q = x^2 + xy + y^2$ has full image for every odd $p \neq 3$. Selecting more good completely split primes supplies no square-type saving for T03. This says nothing by itself about restricted inputs, prime powers, correlated observables or other real carriers. Source: corrected C0, Lemma 4.2 and §10 C3.

Lemma Q4 Prime power norm images and their limits

PROVED in the located norm-trace checkpoint; valuation proof inspected. For $q = x^2 + xy + y^2$ modulo p^k:

- If $p \equiv 1 \pmod 3$, the image is the full ring.
- If $p \equiv 2 \pmod 3$, including $p = 2$, it consists of zero and residues of even valuation. Its size is $(p^{k+1} + 1)/(p + 1)$ for even k, and $(p^{k+1} + p)/(p + 1)$ for odd k. Its density is at least $p/(p + 1)$.
- For $p = 3$, it consists of zero and residues whose first nonzero ternary digit is one. Its size is $(3^k + 1)/2$.

Proof. At a split prime, an invertible local change of variables factors the form into two independent linear forms, so setting one equal to one represents every residue. At an inert prime, anisotropy modulo p forces both inputs divisible by p whenever the output is divisible by p; dividing the form by p^2 gives the even-valuation condition recursively. Every unit is represented by a nonsingular finite-field solution and Hensel lifting. This includes $p = 2$, since at a nonzero input at least one of the partial derivatives y, x is nonzero modulo two. Summing the unit counts at even valuations gives the displayed sizes.

At three, $q \equiv (x - y)^2$. Unit outputs have leading ternary digit one. If the inputs are not both divisible by three and $q \equiv 0$, write $x = y + 3t$; then $q = 3(y^2 + 3yt + 3t^2)$, with valuation one and leading digit one. Common divisibility reduces by nine. Conversely units congruent to one modulo three are 3-adic squares: $q(s, 0) = s^2$ and $q(s, s) = 3s^2$, followed by common powers of three, represent the permitted residues. Summing 3^{k-v-1} over $0 \le v < k$, and adding zero, gives $(3^k + 1)/2$. □

Source: the separate T01 norm-trace checkpoint, §7.1, dated 2026-09-10. This is a local theorem about the homogeneous quadratic, not a solution of its real target or of $P_0 = q + x$. These positive but saturating defects need a separate field-compatibility and packing ledger. Broader asymptotic norm-sieve obstructions use extra arithmetic results and are not conclusions of the original unit-distance paper alone.

EVERY INPUT LOSS AND OUTPUT BRANCH

Appendix C Charged ledgers and scope

These ledgers apply the repository's construction-ledger template to the mechanisms actually described above. A missing support improvement is marked OPEN. Inapplicable costs are explained, rather than silently omitted.

The exact quantitative comparison

Centered rank-one quadratic

Lower input rate

$\log X - \frac{1}{2}\log D$

Full output certificate rate

$\log \rho + \log(2H_2 + Q)$

Gap furnished by the certificate

Positive after the original finite-list parameter choice; Q1-Q2

Side-length cubic rectangle, denominator p

Lower input rate

$\log X - \frac{1}{2}\log R$

Full output certificate rate

$\log \delta_{p,2} + \log[p^2 + 2\max(X^3, p^2 X)]$

Gap furnished by the certificate

$2a - b = -\log \beta_p(X) < 0$; C5-C6

Symmetric T02 fourth-power box

Lower input rate

$\log X - \frac{1}{2}\log D$

Full output certificate rate

$\log \rho_4 + \log(162X^4 + 8Q^3 X + Q)$

Gap furnished by the certificate

$2a - b = -\log \Theta_{4,\mathrm{sym}}(X) < 0$; F2

Ramified-seven C4 carrier

Lower input rate

$\log X - \frac{1}{3}\log 7 - \frac{1}{2}\log D$

Full output certificate rate

$\frac{1}{3}\log N_m + \log(1 + W/7^m)$

Gap furnished by the certificate

Negative for every fixed precision in the specified full-interval certificate; F4-F5

Source norm-one traces

Lower input rate

$\gamma = t \log 2 - \log H$

Full output certificate rate

Existing enclosure: $\log(136 Q_0^6 + 1)$

Gap furnished by the certificate

No saving; the actual output rate is OPEN; N4-N5

All use upper input rates as well: centered boxes use $B = \log(2X + 1)$; the cubic side-length rectangle uses $B = \log(X + 1)$; source traces use $B = t \log 2$. A positive asymptotic gap $\delta = 2a - b$ certifies every strict $c < \delta / B$. Only the exact finite inequalities in Q1-Q2 justify their displayed boundary exponents without extra control.

Where each cost is paid

Input lower and upper counts

Quadratic and cubic additive boxes

G2 or translated averaging; the lower count pays $D^{d/2}$

Symmetric and ramified quartic carriers

G2; C4 ideal lattice also pays $7^{d/3}$ per input

Norm-one trace carrier

Actual ideal-class fiber yields $e^{\gamma d}$; trace injectivity and the binary upper bound give actual scalar counts

Input thinning twice

Quadratic and cubic additive boxes

Zero for full additive boxes; C8 charges a code's density squared

Symmetric and ramified quartic carriers

Zero for F2; C4 charges $7^{-2d/3}$; deeper subsets require a fresh count

Norm-one trace carrier

Class-number loss is already in γ; any additional loss λ changes the pair budget to $2(\gamma - \lambda)$

Field degree and discriminant

Quadratic and cubic additive boxes

Degree grows last; $\log D$ is a cost in the normalized output comparison

Symmetric and ramified quartic carriers

Same, with $d/3$ actual ramified blocks in C4

Norm-one trace carrier

Degree is $[L : \mathbb{Q}]$, not $[L(i) : \mathbb{Q}]$; bounded discriminant enters the retained class-number bound

Local moduli and splitting

Quadratic and cubic additive boxes

G4 gives the actual quotient coordinates; local powers and precision are fixed

Symmetric and ramified quartic carriers

F2 uses split primes; C4 uses ramification index three and relative splitting, not d copies of $\mathbb{F}_7$

Norm-one trace carrier

Original split prime pairs generate valuation labels; no new local output saving is assumed

Denominators and transport

Quadratic and cubic additive boxes

Q, M, and the cleared coefficients occur in the radius and packing modulus; exact common scaling has no fiber loss

Symmetric and ramified quartic carriers

Q^3 occurs in F2's height; C4 with $Q = 1$ is integral

Norm-one trace carrier

Pole denominator Q_0^2 becomes Q_0^6 for cubic outputs; trace embedding is injective, but the enclosure remains expensive

Coefficient height and archimedean distortion

Quadratic and cubic additive boxes

Explicit H_2 or exact coupled cubic width; no unit distortion

Symmetric and ramified quartic carriers

Explicit $81X^4 + 4Q^3X$, or the proved translated width W

Norm-one trace carrier

Every trace conjugate is bounded by two, and outputs by 68; bounded radius does not erase poles

Regulator and class group

Quadratic and cubic additive boxes

Regulator inapplicable and no class selection of these additive carriers

Symmetric and ramified quartic carriers

Inapplicable to the direct integral additive carriers; original arithmetic supply is separately sourced

Norm-one trace carrier

Regulator inapplicable; class-number cost is essential and charged once in input supply

Branches, ratios and carries

Quadratic and cubic additive boxes

Full output classes are counted; C8's correlated code is charged as one actual code

Symmetric and ramified quartic carriers

All C4 local classes and mixed-depth branches are counted; actual characteristic 7^m preserves carries

Norm-one trace carrier

Joint product/ratio labels have quadratic support; the signed branches are attained by actual ordered pairs; their further collapse is OPEN

Parameter and real projection fibers

Quadratic and cubic additive boxes

Zero loss: one real field embedding and scalar multiplication are injective

Symmetric and ramified quartic carriers

Zero loss for the same real embedding and translation/scaling

Norm-one trace carrier

Zero loss for the actual anchored binary trace family; other scalarizations need proved fiber bounds

Exceptions and uncovered pairs

Quadratic and cubic additive boxes

None in the displayed full-square bounds

Symmetric and ramified quartic carriers

None in the displayed full-square bounds

Norm-one trace carrier

None in the enclosure; every exception in a proposed support or tail theorem must be charged

Additional actual-support saving

Quadratic and cubic additive boxes

Established only for the quadratic comparison

Symmetric and ramified quartic carriers

OPEN beyond the retired certificates

Norm-one trace carrier

OPEN; neither selected edges nor energy supplies it

A direct exact-output encoding can avoid paying a full ambient packing capacity. Its cardinality must be proved from actual equalities and cover every scalar ordered pair. Poles and other arithmetic constraints still govern the realization; they do not disappear because a cardinality proof is proposed.

What a replacement theorem must actually say

For the source trace carrier, N5 states the concrete missing bound

$$|P_3(T_d, T_d)| \le C d^m e^{-\kappa d} |T_d|^2, \qquad \kappa > 0,$$

on an unbounded sequence of layers, with all constants fixed. The source upper count then certifies every $c < \kappa/(t \log 2)$. This is a conditional subproblem restricted to that carrier, not an established simplification of T01. Its rich fibers must connect many products and ratios by N3. The existing broader EG-O1/NC-O1 obligation is the actual full-fiber tail in G8; it includes isolated and unprescribed pairs.

For the C4 quartic carrier, if an actual subcarrier loses rate λ, a new branch-dependent all-pairs cover needs total output rate $b_{\text{eff}} < 2(a - \lambda)$. The input loss, every branch, its actual residue population and real width, and every exceptional pair must all enter that inequality. The stabilized local proportion $22/49$ alone supplies no such bound.

For a full-rank quadratic, Q3 blocks unrestricted first-level square-residue compression at good primes. Q4's higher-power densities, a restricted-input norm identity or a solvable-extension realization become useful only after their hypotheses and global costs are proved for one real same-set carrier. Completing a quadratic norm or obtaining many norm-one actions supplies algebra, not automatically a small support.

What the mechanism establishes

Chebotarev selects suitable primes at a finite extension. The original group's relation calculation then enforces their splitting in every layer while keeping the tower infinite. CRT repeats a proved local restriction across actual residue coordinates. Uniform per-class packing controls all arithmetic lifts. The final step is a positive fully charged rate followed by an injective same-set real transport. Each arrow has its own statement above.

The quadratic proof completes every arrow. For the audited cubic and fourth-power routes, the rate comparison fails even though the local defects and tower are valid. A failed numerical certificate is not a lower bound on occupied outputs, a proof of near-quadratic expansion on every real set, or an impossibility theorem by degree. Quartic composition and pure integral-linear powers give explicit positive counterexamples to any degree-only impossibility claim.

The stronger infinite-prime tower in the remarks is useful arithmetic infrastructure, but it does not remove coefficient heights, modulus spacing, squared input losses or full-pair coverage requirements. Likewise, the original unit-distance edge count does not upper-bound the support of a scalar cubic. Its norm-one trace import remains OPEN at the exact output-counting step.

The report uses no sum-product exponent as a polynomial compression exponent. The additive-box/unit-box construction and its published quantitative saving concern a different factorization and cost ledger. The source arithmetic does not print numerical values for all relation-rank and class-number constants or one numerical admissible ℓ; the positive quadratic exponents here are symbolic fixed constants after the parameter choice, not asserted numerical records. No novelty conclusion, independent external review, new mathematical experiment or new construction is claimed.

Source availability

The original unit-distance PDF, corrected C0 with its correction log, the complete cubic checkpoint and continuous-cover proof, and C2-C4 analytic quartic reports are locally present and their relevant arguments were inspected. The original C1 quartic report and some broader recovery-package artifacts are absent from the received local supplement. Stronger C1 assertions are retained as reported state and are not dependencies of the proofs above. The identified external number-theory references behind the original paper have not all been independently read or reproved. These boundaries are recorded in the claim and source ledger and the existing migration conflict register.

The earlier 11 October transfer note recorded Pohoata as unavailable in its offline cache. The present report's source comparison uses the primary v2 text accessed during this conversation. The earlier file remains unchanged; its historical availability claim was true for that task and is not silently rewritten. No theorem is inferred from a missing title or retrospective review acceptance. A later same-date torsion-compatible sharing note and append-only registry updates appeared concurrently. That separate elliptic branch lies outside this report's mechanism audit; the source versions and scopes used here remain explicit, and T01 remains OPEN.

The report is a consolidation and source/algebra audit. Packaging checks validate files and references; they do not prove the research targets. This task creates only this dated report directory. Historical sources, canonical manifests, SPECs, STATUS files and inherited uncommitted work are preserved. The verification record lists the actual checks and their results.

43 INDEXED RECORDS / 23 LOCAL SOURCE IDENTITIES

Appendix D Claim records and source identities

11 October 2026 · version 1

The theorem statements, polynomial identities, quantifiers and hypotheses are in report.md, at the corresponding local claim ID. Every row below is a claim record in this dated report. Its filename and statement locator are `report.md`, the named ID; its version is 1 and date is 2026-10-11. Source versions or byte hashes appear in the source catalogue. Dependencies, evidence, audit performed and unresolved gaps are explicit here. These labels do not reassign original theorem numbers or assert novelty.

The scope is source inspection and exposition of the original arithmetic mechanism, located project proofs and elementary deductions. Network use was limited to the cited primary papers accessed in this conversation. The finite mathematical computation budget for the report is zero: no carrier enumeration, experiment, imported program or proof-search sweep was authorized or run. Documentation and integrity commands are recorded separately. Underlying external arithmetic theorems are retained inputs; internal checking is not external review.

Claim records

A1 / Shared arithmetic

Polynomial or object

None; original unit-distance result

Sources and statement locators

S1

Dependencies and hypotheses

None

Evidence and audit performed

EXTERNAL RESULT; source identity and relevant pages inspected

Remaining gap or exact scope

Underlying cited references not all independently read

A2 / Shared arithmetic

Polynomial or object

Cyclic cubic fields

Sources and statement locators

A1; S1 Prop.3.2

Dependencies and hypotheses

Conductor-discriminant and relative discriminant formulas; PNT in AP

Evidence and audit performed

EXTERNAL RESULT; source proof inspected

Remaining gap or exact scope

External cyclotomic and analytic inputs retained

A3 / Shared arithmetic

Polynomial or object

Pro-3 arithmetic groups

Sources and statement locators

A1; S1 Props.3.3-3.5

Dependencies and hypotheses

Frattini, Shafarevich, Golod-Shafarevich imports

Evidence and audit performed

EXTERNAL RESULT; exact forms and use inspected

Remaining gap or exact scope

No independent proof of the external group theorems

A4 / Shared arithmetic

Polynomial or object

Finite splitting data

Sources and statement locators

A2-A3; S1 Prop.3.6

Dependencies and hypotheses

Chebotarev in a finite normal closure

Evidence and audit performed

EXTERNAL RESULT; choice and exclusions inspected

Remaining gap or exact scope

No arbitrary preassigned prime list or effective prime-size bound

A5 / Shared arithmetic

Polynomial or object
Totally real tower

Sources and statement locators
A2-A4; S1 Prop.3.7-3.8 and Remark3.9

Dependencies and hypotheses
Class-number estimate; finite Frobenius quotient

Evidence and audit performed
EXTERNAL RESULT; assembly and parameter order inspected

Remaining gap or exact scope
Infinite common prime set is not this statement

A6 / Shared norm-one directions

Polynomial or object
Relative norm one in L(i)/L

Sources and statement locators
A5; S1 Prop.2.2

Dependencies and hypotheses
Ideal-class pigeonhole and valuation labels

Evidence and audit performed
EXTERNAL RESULT; argument inspected

Remaining gap or exact scope
Directions have poles; no scalar polynomial support theorem

A7 / Shared selected translations

Polynomial or object
Planar unit distances

Sources and statement locators
A6; S1 Thm.2.3 and Lemmas2.4-2.6

Dependencies and hypotheses
Overlap averaging, faithful complex embedding, norm packing

Evidence and audit performed
EXTERNAL RESULT; proof chain inspected

Remaining gap or exact scope
Only selected edges; no full scalar square count

A8 / Shared optional arithmetic

Polynomial or object
Real scalarization of directions

Sources and statement locators
A5-A7; W2 Prop.2.3

Dependencies and hypotheses
Recursive HMR cutting for optional infinite-prime input

Evidence and audit performed
EXTERNAL RESULT plus elementary scalarization deductions

Remaining gap or exact scope
HMR underlying theorem not independently checked; scalar support OPEN

G1 / Shared support counting

Polynomial or object
All integral polynomial outputs

Sources and statement locators
Integral norm; S2 §5.3

Dependencies and hypotheses
Totally real signature and integral differences

Evidence and audit performed
PROVED; complete elementary proof reproduced

Remaining gap or exact scope
Does not give separation in one chosen real embedding

G2 / Shared support counting

Polynomial or object
All integral carriers

Sources and statement locators
G1; S2 §5.2

Dependencies and hypotheses
Minkowski covolume and Blichfeldt averaging

Evidence and audit performed
PROVED; complete elementary proof reproduced

Remaining gap or exact scope
Must fix X above sqrt(D) for growth

G3 / Shared support counting

Polynomial or object
All integral polynomial outputs

Sources and statement locators
G1; S2 §5.3

Dependencies and hypotheses
Uniform maximum-norm packing

Evidence and audit performed
PROVED; complete elementary proof reproduced

Remaining gap or exact scope
An enclosure need not be filled

G4 / Shared support counting

Polynomial or object
Actual finite quotient restrictions

Sources and statement locators
CRT; S4 §E.1

Dependencies and hypotheses
Complete splitting at the stated rational primes

Evidence and audit performed
PROVED; quotient argument inspected and reproduced

Remaining gap or exact scope
Other decomposition types require their actual ledger

G5 / Shared support gate

Polynomial or object

Fixed F on centered boxes

Sources and statement locators

G1-G4

Dependencies and hypotheses

Entire-image radius and residue containment

Evidence and audit performed

PROVED implication; complete calculation reproduced

Remaining gap or exact scope

Positive rate required; failing it gives no image lower bound

G6 / Shared rate gate

Polynomial or object

Fixed P on actual real sets

Sources and statement locators

S15 A3

Dependencies and hypotheses

Lower and upper input rates; full output rate

Evidence and audit performed

PROVED implication; complete proof reproduced

Remaining gap or exact scope

Strict exponent only with unspecified o(d) errors

G7 / Shared local algebra

Polynomial or object

Common-scaling powers and T01/T02 reductions

Sources and statement locators

S2 §4.1; S4 §E.1; F7; W1

Dependencies and hypotheses

Homogeneous decomposition and cyclic unit groups

Evidence and audit performed

PROVED; complete elementary derivations reproduced

Remaining gap or exact scope

Exceptional primes and correlated inputs need separate statements

G8 / Shared support bookkeeping

Polynomial or object

Fixed P on same-set finite inputs

Sources and statement locators

S15 A3; S16 C5

Dependencies and hypotheses

Actual input and output sets or exact scalar fibers

Evidence and audit performed

PROVED; complete identities reproduced

Remaining gap or exact scope

No automatic real realization, tensorization or useful tail

Q1 / Shared rank-one quadratic

Polynomial or object

Integral L^2+T with stated primitivity/transversality

Sources and statement locators

A5; G1-G7; S2 Thm.C0.1; S3

Dependencies and hypotheses

All finite parameters fixed; full box

Evidence and audit performed

PROVED in source system; complete argument inspected/reproduced

Remaining gap or exact scope

No novelty or general full-rank theorem

Q2 / Quadratic reference

Polynomial or object

$(x-y)^2+x+y$

Sources and statement locators

A5 or optional A8; G1-G7; W1 Thm.1.1/Prop.4.1; S15 A2

Dependencies and hypotheses

13D rho<1; X fixed after Q

Evidence and audit performed

PROVED elementary reconstruction and EXTERNAL RESULT comparison

Remaining gap or exact scope

Original unit-distance theorem itself does not state polynomial result

Q3 / Full-rank quadratic local algebra

Polynomial or object

Nondegenerate binary q; T03 $q=x^2+xy+y^2$

Sources and statement locators

S2 Lemma4.2

Dependencies and hypotheses

Odd good prime; rank condition

Evidence and audit performed

PROVED; finite-field proof inspected/reproduced

Remaining gap or exact scope

Restricted sets and higher powers remain separate

Q4 / Full-rank quadratic local algebra

Polynomial or object

x^2+xy+y^2 modulo p^k

Sources and statement locators

S7 §7.1

Dependencies and hypotheses

Split/inert/ramified local norm classification

Evidence and audit performed

PROVED; valuation and nonsingular-lifting argument inspected

Remaining gap or exact scope

Global costs and real witness not supplied

Q5 / Quadratic interval comparison

Polynomial or object

$f=(x-y)^2+x+y$

Sources and statement locators

G7; W1 Prop.2.1/Cor.2.2

Dependencies and hypotheses

Squarefree Q; PNT for asymptotic prime choice

Evidence and audit performed

PROVED interval argument; EXTERNAL RESULT asymptotic comparison

Remaining gap or exact scope

This comparison yields a vanishing saving exponent

C1 / T01 local cubic

Polynomial or object

$P_C=(x-y)^3+C(x+y)$

Sources and statement locators

S4 E1

Dependencies and hypotheses

Odd p; exact valuation depth r

Evidence and audit performed

PROVED; complete local image/fiber proof reproduced

Remaining gap or exact scope

p=2 outside this statement

C2 / T01 local cubic

Polynomial or object

Cube residues modulo p^r

Sources and statement locators

C1; S4 E1

Dependencies and hypotheses

Odd p not equal to3; specified congruence for defect

Evidence and audit performed

PROVED; valuation-shell count reproduced

Remaining gap or exact scope

At3 use separately stated exceptional calculation

C3 / T01 split amplification

Polynomial or object

Cleared $P3=(x-y)^3+x+y$

Sources and statement locators

C1-C2; G4; S4 E1

Dependencies and hypotheses

Complete splitting and usable coefficient depth

Evidence and audit performed

PROVED; quotient and saturation argument reproduced

Remaining gap or exact scope

No density tending to zero by extra depth

C4 / T01 continuous width

Polynomial or object

$Z_M=(u-v)^3+M^2(u+v)$

Sources and statement locators

S4 E2/H4

Dependencies and hypotheses

One positive-sign interval of side X

Evidence and audit performed

PROVED; complete coupled range proof reproduced

Remaining gap or exact scope

Different signs and arbitrary sparse supports not included

C5 / T01 global cubic certificate

Polynomial or object

$P3=(x-y)^3+x+y$

Sources and statement locators

C1-C4; G1/G3; S4 D2/E3/F1

Dependencies and hypotheses

Translated rectangular input and all allowed lifts

Evidence and audit performed

PROVED upper bound; complete count reproduced

Remaining gap or exact scope

A further actual-support improvement is not supplied

C6 / T01 optimization

Polynomial or object

P3 via one good split prime

Sources and statement locators

C2/C5; S4 D1/F2

Dependencies and hypotheses

p=1 mod3; recorded refinement1 mod12

Evidence and audit performed

PROVED minimization; FAILED certificate

Remaining gap or exact scope

No actual-image lower bound

C7 / T01 multi-prime/depth

Polynomial or object

P3; rectangular/algebraic-denominator variants

Sources and statement locators

C2-C5; S4 F2/F3/H4

Dependencies and hypotheses

Good primes; exact denominator/modulus/window hypotheses

Evidence and audit performed

PROVED scoped obstruction and exceptional tests inspected

Remaining gap or exact scope

Other actual-support mechanisms remain OPEN

C8 / T01 correlated input certificate

Polynomial or object

P3 with first-level code

Sources and statement locators

C4-C5; S4 E4

Dependencies and hypotheses

Averaged code-density input count and uniform allowed lifts

Evidence and audit performed

PROVED; complete twice-charged calculation reproduced

Remaining gap or exact scope

Anomalously dense realization plus refined lift not excluded

C9 / T01 positive-volume cover

Polynomial or object

Coordinatewise P3 on W×W

Sources and statement locators

S5 F3/G1

Dependencies and hypotheses

Bounded Borel W of positive volume; full continuous image cover

Evidence and audit performed

PROVED; Jacobian and Cauchy-Schwarz proof reproduced

Remaining gap or exact scope

No lower bound on sparse finite lattice image

N1 / T01 norm-one trace carrier

Polynomial or object

P3 on tau(U)

Sources and statement locators

A6; S4 H2; S6 §4.5

Dependencies and hypotheses

Actual anchored binary ideal-class family

Evidence and audit performed

PROVED deductions inspected; marginal product count retained as reported

Remaining gap or exact scope

No arbitrary real projection guarantee

N2 / T01 trace label identities

Polynomial or object

P3 on tau(U)

Sources and statement locators

N1; S4 H2

Dependencies and hypotheses

Both product and ratio traces, actual signed branches

Evidence and audit performed

PROVED; identities and at-most8 joint fibers reproduced

Remaining gap or exact scope

Joint label support is already quadratic

N3 / T01 fixed-label cubic fibers

Polynomial or object

P3 on tau(U)

Sources and statement locators

N1-N2; S4 H2

Dependencies and hypotheses

One exact product or ratio plus an exact output

Evidence and audit performed

PROVED; nonzero degree-at-most6 equation reproduced

Remaining gap or exact scope

Full cubic fibers remain OPEN

N4 / T01 trace enclosure

Polynomial or object

P3 on tau(U)

Sources and statement locators

N1; G1; S4 H2

Dependencies and hypotheses

Q_0^2 trace poles; entire radius68

Evidence and audit performed

PROVED enclosure; FAILED numerical certificate

Remaining gap or exact scope

Actual trace image size unresolved

N5 / T01 missing support premise

Polynomial or object

P3 on source trace family or general real A

Sources and statement locators

N1-N4; G6/G8; S15-S17

Dependencies and hypotheses

Entire support or actual near-full large-fiber tail

Evidence and audit performed

PROVED implications; OPEN premises

Remaining gap or exact scope

No source carrier satisfies them in the current record

F1 / T02 fourth-power local algebra

Polynomial or object

P4=(x+2y)^4+3x-y

Sources and statement locators

A5; G4/G7; S2 §10 C2

Dependencies and hypotheses

Selected p=1 mod4; exact Q^3 coefficient

Evidence and audit performed

PROVED algebra/count inspected and reproduced

Remaining gap or exact scope

Local restriction alone is insufficient

F2 / T02 symmetric certificate

Polynomial or object

P4 on normalized centered boxes

Sources and statement locators

F1; G1-G5; S2 §10 C2/S3

Dependencies and hypotheses

One valid symmetric enclosure and uniform lifts

Evidence and audit performed

PROVED obstruction; FAILED certificate

Remaining gap or exact scope

Sharper actual-support counts remain OPEN

F3 / T02 C4 local theorem

Polynomial or object

P4 on full critical inverse images

Sources and statement locators

F7 for first-level context; S13 §§1-4

Dependencies and hypotheses

Actual e=3,f=1 valuation ring; all m≥1; Q=1

Evidence and audit performed

PROVED in source system; complete analytic support proof reproduced

Remaining gap or exact scope

No optimum over arbitrary deeper input subsets

F4 / T02 C4 rate

Polynomial or object

P4 on -1+B_I(X)

Sources and statement locators

A2-A3 adapted; F3; G1-G6; S11 §§7-8; S10/S12; S13 §5

Dependencies and hypotheses

Base prime7 split only in relative layers; fixed radius/precision

Evidence and audit performed

PROVED carrier and counts; FAILED positive-rate certificate

Remaining gap or exact scope

No branch-dependent improved actual support supplied

F5 / T02 full-support cover retirement

Polynomial or object

P4 with7-supported ideal-coset covers

Sources and statement locators

F3-F4; S13 §6; S14

Dependencies and hypotheses

Entire local product support; full intervals; uniform norm packing

Evidence and audit performed

PROVED scoped envelope obstruction inspected/reproduced

Remaining gap or exact scope

Numerical envelope lower bounds are not actual-image lower bounds

F6 / Separate quartic/pure-power positives

Polynomial or object

g composed with quadratic; integral L^k

Sources and statement locators

Q1-Q2; direct image inclusion; S19/S18 for C1 availability

Dependencies and hypotheses

Fixed univariate g or integral L; same A

Evidence and audit performed

PROVED elementary consequences; C1 statement only reported

Remaining gap or exact scope

T02 distinct; unavailable C1 proof not load-bearing

F7 / T02 C2/C3 local optimum

Polynomial or object

F_Q on $F_7[\pi]/\pi^3$

Sources and statement locators

S9 §5; S10/S12; S11 §§2-5

Dependencies and hypotheses

Every nonempty subset, $s=\pm1$; normalized input loss twice

Evidence and audit performed

PROVED in source system; complete analytic case argument reproduced

Remaining gap or exact scope

One-level optimum does not imply higher-level optimum

F8 / T02 C1 and correlated inputs

Polynomial or object

F_Q and coordinatewise fourth powers

Sources and statement locators

S8 §4 C1/§5.1; S19 for missing main proof; S23 §§1-2

Dependencies and hypotheses

Exact interval and input-subset hypotheses in report; $p=5,d=2$ case independent of missing main proof

Evidence and audit performed

C1 results PROVED as reported; available finite-case proof inspected/reproduced; general correlated premise OPEN

Remaining gap or exact scope

Missing C1 main proof is not load-bearing for F2-F5; finite case is not an asymptotic theorem

Located source catalogue

Paths are relative to the repository root. Private source copies are not embedded in this report and are not included in an ordinary Git clone. Their hashes identify the inspected local version; hashes are not proof certificates. The supplied archive and both canonical manifests remain unchanged.

S1

Exact source locator

`local-only/migration/research_supplement/shared/03-unit-distance-proof.pdf`

Recorded version and relevant sections

No printed date or version; Original §§2-3, Appendix A; printed pp.6-16

SHA-256 of inspected local file

`62b5de8cacdfbc0f13ae1940aa81659cd233e6945b936612fbeab0ab8ab3d383`

S2

Exact source locator

`local-only/migration/research_supplement/shared/Phase_C0_Corrected.md`

Recorded version and relevant sections

2026-09-03 corrected version; Theorem C0.1; §§4-6; §10 C2/C3

SHA-256 of inspected local file

`8ef3670bb8903b2f851b1bba594435d8635882849dae37912baadf6c49eb85be`

S3

Exact source locator

`local-only/migration/research_supplement/shared/C0_CORRECTION_LOG.md`

Recorded version and relevant sections

Correction chain for S2; Field signature, radius and certificate-scope corrections

SHA-256 of inspected local file

`5da48e6372e1ea10af8d395c30faff6e7676bc546749ffa5fc36765a7b040480`

S4

Exact source locator

`research/history-sources/Cubic_Difference_Sum_Research_Checkpoint.md`

Recorded version and relevant sections

2026-09-02; §§B, D-F, H.2/H.4

SHA-256 of inspected local file

`1a5ecffb8366cd4ef2a0447d16bc183c7c8cde646428db5bc6163959d1e95b92`

S5

Exact source locator

`research/history-sources/Concrete_Cubic_Compression_Block_NoGo_Checkpoint.md`

Recorded version and relevant sections

2026-09-03; §§F.3, G.1; positive-volume cover theorem

SHA-256 of inspected local file

`413dcc5aa06ddc03129d76a68360dc1737a0a7d156d70e27a510a4699d5bbc62`

S6

Exact source locator

`local-only/migration/sources/T01_CURRENT_STATE.md`

Recorded version and relevant sections

2026-09-10 export; §§4.5, 6.1-6.7, 12-15

SHA-256 of inspected local file

`2095cb7b4a5f7d848dea1356956ad36f5303e4037b13b2efe7fa32b0c08f2624`

S7

Exact source locator

`local-only/migration/sources/T01_Norm_Trace_Quadratic_Research_Checkpoint.md`

Recorded version and relevant sections

2026-09-10; §§1-2 and 7.1; separate P0 branch

SHA-256 of inspected local file

`a9de1edb1ebc43465de7e12cd2e815e8647b4199a0c303afff2e4a8815abc0db`

S8

Exact source locator

`local-only/migration/sources/T02_CURRENT_STATE.md`

Recorded version and relevant sections

2026-09-10 export; C2-C4 statements; §§17-20

SHA-256 of inspected local file

`e4ba2cbf3871d71ac45526fa7b16a0f6e468f13cce13261adafec5547830f480`

S9

Exact source locator

`local-only/migration/research_supplement/T02/phase_c2/Phase_C2_Hybrid_Carrier_Structural_Lemma.md`

Recorded version and relevant sections

Reviewed source version identified by hash; §5 determinant graph and local reduction

SHA-256 of inspected local file

`3d2171e450c691f64cc6f25778e53ebb40a8b829e9b470552043aa67fe5264d9`

S10

Exact source locator

`local-only/migration/research_supplement/T02/phase_c3/C2_PRECISION_ADDENDUM.md`

Recorded version and relevant sections

Reviewed correction chain; Parameter support versus actual support and determinant charge

SHA-256 of inspected local file

`c747922830646d3311cb106ac2b11b7ee80e30e3a8c99696df71bfa0d378d032`

S11

Exact source locator

`local-only/migration/research_supplement/T02/phase_c3/Phase_C3_Ramified_7_Adic_Input_Cost.md`

Recorded version and relevant sections

Reviewed source version identified by hash; §§2-5 analytic optimum; §§7-8 tower, ideal count and width

SHA-256 of inspected local file

`50f80f0a203ccff482560f8ed2aa95123501c13fb11440be1526f6048ee712af`

S12

Exact source locator

`local-only/migration/research_supplement/T02/phase_c3/C2_C3_REVIEW_CHANGE_LOG.md`

Recorded version and relevant sections

Reviewed correction chain; Q=1 width and scope corrections

SHA-256 of inspected local file

`016afcd90e64b0d22f5267e3f1f05960465baacf19cf080687bfeebd0a0dc901`

S13

Exact source locator

`local-only/migration/research_supplement/T02/phase_c4/Phase_C4_Higher_Ramified_Precision.md`

Recorded version and relevant sections

Accepted freeze 2026-09-10; §§1-4 local analytic theorem; §§5-6 charged rates and covers

SHA-256 of inspected local file

`0c438ad31f11c48ff8a74bff5454bf1eda8dd7cc3b5f8b2cd05e7dd6b582a91c`

S14

Exact source locator

`local-only/migration/sources/T02_C4_REVIEW_ACCEPTANCE_AND_HANDOFF(1).md`

Recorded version and relevant sections

2026-09-10; Accepted state and exact retirement scope

SHA-256 of inspected local file

`8d093d4111e04dfd42d6bf13117305809beffceb44f1e8d81f16b64d4ef7c714`

S15

Exact source locator

`research/notes/2026-10-11-transfer-division-entropy/A_transfer.md`

Recorded version and relevant sections

2026-10-11 version 1; A2-A5; baseline and rate ledger

SHA-256 of inspected local file

`0eb275050c0f36881bc0bea52f9474aa7c713d1ec43e4507effaffbd90f98bce`

S16

Exact source locator

`research/notes/2026-10-11-transfer-division-entropy/C_entropy.md`

Recorded version and relevant sections

2026-10-11 version 1; C3 covering bridge; C5 support/fiber identities

SHA-256 of inspected local file

`0ecfe1d4d4ea7682d1e247becf4697a07a49de1b47a89558550ef54760f15af8`

S17

Exact source locator

`research/notes/2026-10-11-transfer-division-entropy/README.md`

Recorded version and relevant sections

2026-10-11 version 1; T01 transfer record and its actual fiber-tail obligation

SHA-256 of inspected local file

`4d644f2926627bb70f94f1735fec44f87a19ac489905c165c2f7b79285657e7d`

S18

Exact source locator

`research/MIGRATION_CONFLICTS.md`

Recorded version and relevant sections

2026-09-11 with 2026-09-20 addenda; Source hierarchy; MC-01, MC-06 and registry

SHA-256 of inspected local file

`f1524d8415086650f8902eb2df8fb3c356517db6575ecbcf648b60ff4e9bd37a`

S19

Exact source locator

`local-only/migration/recovery_2026-09-20/T02/T02_RECOVERY_ADDENDUM_2026-09-20.md`

Recorded version and relevant sections

2026-09-20; mathematical cutoff 2026-09-10; Unavailable C1 main proof and package boundary

SHA-256 of inspected local file

`c10dbc1cea20ad3b30bc721f73261a456871217b4499c083f0f15f5b07dadb70`

S20

Exact source locator

`local-only/migration/sources/00_MASTER_HANDOFF.md`

Recorded version and relevant sections

2026-09-11; Targets, source discipline and gain-cost framework

SHA-256 of inspected local file

`b5ea32664e1a755faa4b29ad21848e11f3b16eef94b451d5b3ed5e9d18f6b250`

S21

Exact source locator

`local-only/migration/sources/T03_CURRENT_STATE.md`

Recorded version and relevant sections

2026-09-10 export; Full-rank target and scoped norm-sieve history

SHA-256 of inspected local file

`8048b4d9d3121757a28abe90e227fa5fba84cb4eb172bef8b6aa7186b3792850`

S22

Exact source locator

`local-only/migration/sources/T04_CURRENT_STATE.md`

Recorded version and relevant sections

2026-09-10 export; Distinct negative-linear cubic and source boundaries

SHA-256 of inspected local file

`17d5bb792fc4a9903bac337e0b96d283bad551796ddccaa3b8965816b8ac59b5`

S23

Exact source locator

`local-only/migration/research_supplement/T02/phase_c2/C1_STATUS_ADDENDUM.md`

Recorded version and relevant sections

Reviewed correction version identified by hash; §1 weak deficit, §2 complete p=5,d=2 proof

SHA-256 of inspected local file

`adb1d2761dcd983d1fc6e33acac1fcee68c8839cf9b48056892c2604d370c556`

Primary web comparisons

- **W1:** Cosmin Pohoata, *Split primes and the Elekes-Rónyai problem*, arXiv:2606.13619v2, revised 28 June 2026. Relevant statements: Theorem 1.1; Proposition 2.1 and Corollary 2.2; Proposition 3.1; Lemmas 3.2-3.3; Proposition 4.1 and common scaling. Primary text accessed during this conversation. No local PDF byte hash is asserted. Q1-Q2 can use S1's finite-list tower instead of W1's stronger input.
- **W2:** Alon, Bloom, Gowers, Litt, Sawin, Shankar, Tsimerman, Wang and Matchett Wood, *Remarks on the Disproof of the Unit Distance Conjecture*, official PDF, Proposition 2.3, printed p.6. The infinite common split-prime statement and its recursive Chebotarev explanation were inspected. The underlying HMR theorem was not independently reproved or assigned a new source label. No printed version date or local hash is fabricated.
- The official original unit-distance PDF was accessed for the original statement locators. S1 identifies the local supplied copy. No byte-identity claim between the web response and S1 is made.

Source limitations and correction chain

S3 controls field-signature, valid-radius and quartic-certificate wording in S2. S10 and S12 control C2's determinant, parameter support, generator-elimination and Q=1 scope. S14 fixes the accepted C4 retirement scope. The source chronology, including unavailable C1 originals, is retained in S18-S19. No missing formal proof was reconstructed from its title, summaries, acceptance or finite data.

The report does not use a universal full-rank quadratic real theorem, a universal cubic or quartic impossibility theorem, arbitrary preassigned-prime splitting, or a theorem that every higher-precision input subset has C4's normalized factor. It uses no sum-product exponent as a saving for T01 or T02. The trace support premise in N5 and the realized-support quartic premise remain OPEN. T03 and T04 state files S21-S22 establish target identities and recorded OPEN status; their entire later research histories are not independently audited here.

PDF PRESENTATION

Production scope

This PDF presents the 11 October 2026 version-1 report in a results-first order. It adds an opening brief and reading guide, then preserves the mathematical statements, proofs, source distinctions and unresolved gaps. No new mathematical search, experiment or proof result is introduced by PDF production.

The Markdown report and claim ledger remain the underlying records. Source paths and hashes identify the inspected local versions; hashes and file checks are not mathematical proofs. The final PDF is checked for mathematical rendering errors, layout overflow and legibility.